

NURTURE COURSE
FLUID MECHANICS

CONTENTS

FLUID MECHANICS

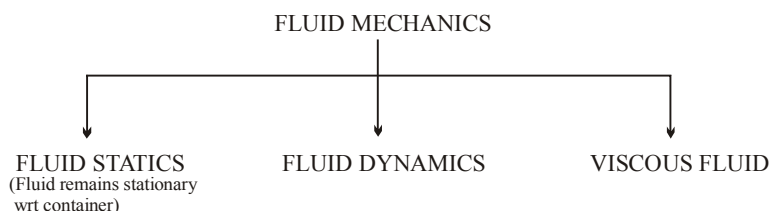
• Key-Concepts	: Formulae and Basic Theory	1
• Exercise (S-1)	: Conceptual Subjective Problems	39
• Exercise (S-2)	: Challenging Subjective Problems	44
• Exercise (O-1)	: Miscellaneous Type Problems	50
• Exercise (O-2)	: Miscellaneous Type Problems	61
• Exercise (J-M)	: Previous 10 years AIEEE Problems	71
• Exercise (J-A)	: Previous 10 years IIT-JEE Problems	74
• Answer Key		80

FLUID MECHANICS

KEY CONCEPTS

What Is a Fluid?

A **fluid**, in contrast to a solid, is a substance that can flow. Fluids conform to the boundaries of any container in which we put them. They do so because a fluid cannot sustain a force that is tangential to its surface. A fluid is a substance that flows because it cannot withstand a shearing stress. It can, however, exert a force in the direction perpendicular to its surface.



Fluid includes property → (A) Density (B) Viscosity (C) Bulk modulus of elasticity (D) Pressure (E) Specific gravity.

Assumptions used in fluid mechanics

1. Fluid is incompressible means density remains constant and volume also remains constant.
2. Fluid is non viscous. There is no tangential force between two layers.

DENSITY (ρ)

Mass per unit volume is defined as density. So density at a point of a fluid is represented as

$$\rho = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V} = \frac{dm}{dV} \quad \text{Density is a positive scalar quantity.}$$

SI UNIT : kg/m^3

CGS UNIT : g/cc

Dimensions : $[\text{ML}^{-3}]$

RELATIVE DENSITY

It is defined as the ratio of the density of the given fluid to the density of pure water at 4°C .

$$\text{Relative density (R.D.)} = \frac{\text{density of given liquid}}{\text{density of pure water at } 4^\circ\text{C}}$$

Relative density or specific gravity is a unitless and dimensionless positive scalar physical quantity. Being a dimensionless/unitless quantity R.D. of a substance is same in SI and CGS system.

SPECIFIC GRAVITY

It is defined as the ratio of the specific weight of the given fluid to the specific weight of pure water at 4°C .

$$\text{Specific gravity} = \frac{\text{specific weight of given liquid}}{\text{specific weight of pure water at } 4^\circ\text{C} (9.81 \text{ kN/m}^3)} = \frac{\rho_\ell \times g}{\rho_w \times g} = \frac{\rho_\ell}{\rho_w} = \text{R.D. of the liquid}$$

Thus specific gravity of a liquid is numerically equal to the relative density of that liquid and for calculation purposes they are used interchangeably.

Solution :

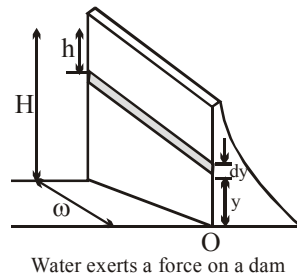
The torque is $\tau = \int d\tau = \int r dF$

From the figure $\tau = \int_0^H y[\rho g(H-y)w dy] = \frac{1}{6} \rho g w H^3$

The total force is given as $\frac{1}{2} \rho g w H^2$

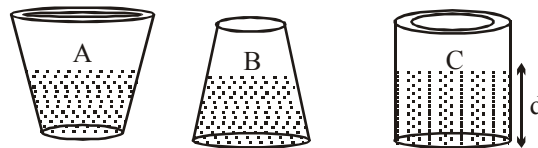
If this were applied at a height y_{eff} such that the torque remains unchanged, we have

$$\frac{1}{6} \rho g w H^3 = y_{\text{eff}} \left[\frac{1}{2} \rho g w H^2 \right] \quad \text{and} \quad y_{\text{eff}} = \frac{1}{3} H$$



Example(a&b) :

Three vessels having different shapes are as shown in the figure below, they have same base area and the same weight when empty (Fig.). The vessels are filled with mercury to the same level. Neglect the effect of the atmosphere. (a) Which have the largest and which have the smallest pressures at the bottom of the vessel or are they same? (b) Which show the highest weight when weighed on a weighing scale or are they same?



Three differently shaped vessels filled with water to same level.

Solution (a) The mercury at the bottom of each vessel is at the same depth d below the surface. Neglecting the pressure at the surface, the pressures at the bottom must be equal hence:

$$P = \rho g d$$

(b) The weight of each filled vessel is equal to the weight of the vessel itself plus the weight of the mercury inside. The vessels themselves are of equal weights, but vessel A holds more mercury than C, while vessel B holds less mercury than C. Vessel A weighs the most and vessel B weighs the least.

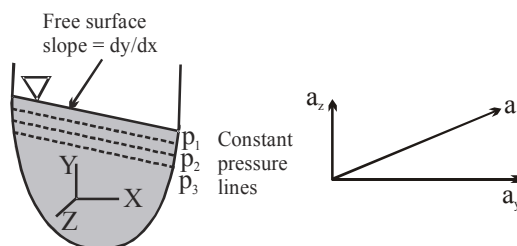
Example(c) : As the mercury exerts the same downward force on the bottom of each vessel, then why does the vessels weigh differently ?

Soln. In vessel C, forces due to fluid pressure on the sides of the container are horizontal. Forces on any two diametrically opposite points on the walls of the container are equal and opposite; thus, the net force on the container walls is zero. The force on the bottom is

$$F = PA = (\rho g d) (\pi r^2)$$

The volume of water in the cylinder is $V = \pi r^2 d$, so

$$F = \rho g V = (\rho V) g = mg$$



Along a free surface the pressure is constant, so that for the accelerating mass shown in Fig. () the free surface will be inclined if $a_x \neq 0$. In addition, all lines parallel to the free surface will have same pressure. For the special circumstance in which $a_x = 0$, $a_y \neq 0$, which corresponds to the mass of fluid accelerating in the vertical direction, Eq. () indicates that the fluid surface will be horizontal. However, from Eq. () we see that the pressure variation is not $\rho g dy$, but is given by the equation.

$$dP = \rho(g + a_y)dy$$

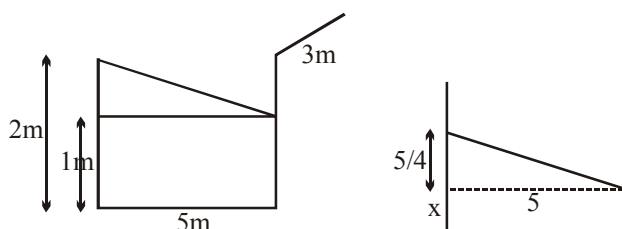
Thus, the pressure on the bottom of a liquid-filled tank which is resting on the floor of an elevator that is accelerating upward will be more than, if the, tank would have been at rest (or moving with a constant velocity). It is to be noted that for a freely falling fluid mass ($a_y = -g$), the pressure variation in all three coordinate directions are zero, which means that the pressure throughout will be same. The pressure throughout a "blob" of a liquid floating in an orbiting space shuttle (a form of free fall) is zero. The only force holding the liquid together is surface tension see Chapter().

Ques : The cross section of a tank kept on a vehicle is shown in Fig.. The rectangular tank is open to the atmosphere. During motion of the vehicle, the tank is subjected to a constant linear acceleration, $a = 2.5 \text{ m/s}^2$. How much fluid will be left inside the tank if initially the tank is half filled. The vessel is 5m wide and 2m high.

Ans.

If the height of the liquid on the left wall is greater than 2m the fluid will be spilled out. Using equation () we can find the angle that the fluid will make with the horizontal.

$$\tan \theta = \frac{2.5}{10} = \frac{1}{4}$$



Lets assume that the dimension of tank in the plane perpendicular to the page is d .

From the geometry its easy to see that free surface on RHS will go down and will rise on LHS. Thus if we assume that fluid on RHS has not touched the floor, we will have fluid taking a shape as described in the diagram. The cuboid part will have volume $x \times 5 \times d$, where x is the height above the bottom.

The wedge part will have the volume $\frac{1}{2} \times h \times 5 \times d$ where h can be found as following

$$(h/5) = \tan \theta = (1/4)$$

Demonstrating Pascal's Principle

Consider the case in which the incompressible fluid is a liquid contained in a tall cylinder, as in Fig. The cylinder is fitted with a piston on which a container of lead shot rests. The atmosphere, container, and shot exert pressure p_{ext} on the piston and thus on the liquid. The pressure p at any point P in the liquid is then

$$p = p_{\text{ext}} + \rho gh$$

Let us add a little more lead shot to the container to increase p_{ext} by an amount Δp_{ext} . The quantities Δp_{ext} , g and h in Eq. are unchanged, so the pressure change at P is

$$\Delta p = \Delta p_{\text{ext}}$$

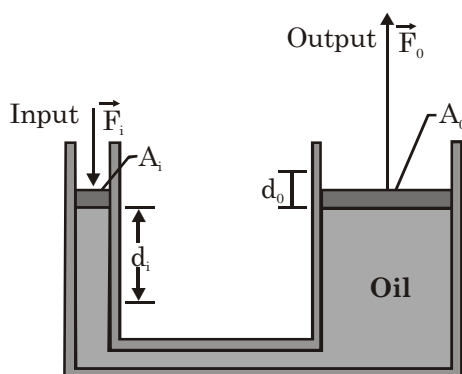
This pressure change is independent of h , so it must hold for all points within the liquid, as Pascal's principle states.

Pascal's Principle and the Hydraulic Lever

Figure shows how Pascal's principle can be made the basis of a hydraulic lever. In operation, let an external force of magnitude F_i be directed downward on the left-hand (or input) piston, whose surface area is A_i . An incompressible liquid in the device then produces an upward force of magnitude F_o on the right-hand (or output) piston, whose surface area is A_o . To keep the system in equilibrium, there must be a downward force of magnitude F_o on the output piston from an external load (not shown). The force $\vec{F}_i$ applied on the left and the downward force $\vec{F}_o$ from the load on the right produce a change Δp in the pressure of the liquid that is given by

$$\Delta p = \frac{F_i}{A_i} = \frac{F_o}{A_o}$$

$$F_o = F_i \frac{A_o}{A_i}$$



Equation shows that the output force F_o on the load must be greater than the input force F_i if $A_o > A_i$ as is the case in figure.

If we move the input piston downward a distance d_i , the output piston moves upward a distance d_o , such that the same volume V of the incompressible liquid is displaced at both pistons. Then

$$V = A_i d_i = A_o d_o$$

which we can write as

$$d_o = d_i \frac{A_i}{A_o}$$

This shows that, if $A_o > A_i$ (as in Figure), the output piston moves a smaller distance than the input piston moves.

From Eqs. we can write the output work as

$$W = F_o d_o = \left(F_i \frac{A_i}{A_o} \right) \left(d_i \frac{A_i}{A_o} \right) = F_i d_i$$

which shows that the work W done on the input piston by the applied force is equal to the work W done by the output piston in lifting the load placed on it.

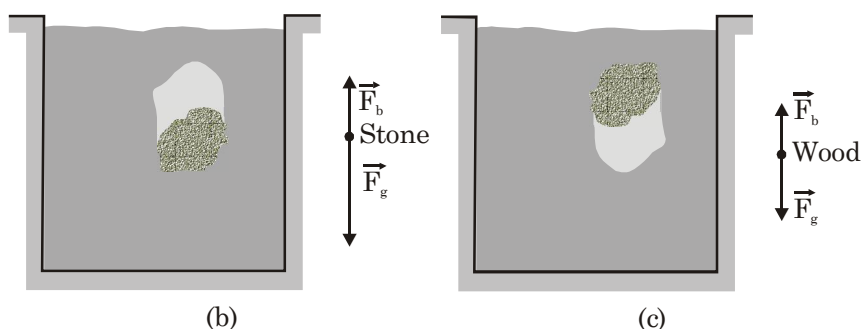
The advantage of a hydraulic lever is this:

With a hydraulic lever, a given force applied over a given distance can be transformed to a greater force applied over a smaller distance.

The product of force and distance remains unchanged so that the same work is done. However, there is often tremendous advantage in being able to exert the larger force. Most of us, for example, cannot lift an automobile directly but can with a hydraulic jack, even though we have to pump the handle farther than the automobile rises and in a series of small strokes.

Archimedes' Principle

Figure shows a student in a swimming pool, manipulating a very thin plastic sack (of negligible mass) that is filled with water. She finds that the sack and its contained water are in static equilibrium, tending neither to rise nor to sink. The downward gravitational force $\vec{F}_g$ on the contained water must be balanced by a net upward force from the water surrounding the sack.



This net upward force is a **buoyant force** $\vec{F}_b$. It exists because the pressure in the surrounding water increases with depth below the surface. Thus, the pressure near the bottom of the sack is greater than the pressure near the top, which means the forces on the sack due to this pressure are greater in magnitude near the bottom of the sack than near the top. Some of the forces are represented in Fig. (a), where the space occupied by the sack has been left empty. Note that the force vectors drawn near the bottom of that space (with upward components) have longer lengths than those drawn near the top of the sack (with downward components). If we vectorially add all the forces on the sack from the water, the horizontal components cancel and the vertical components add to yield the upward buoyant force $\vec{F}_b$ on the sack. (Force $\vec{F}_b$ is shown to the right of the pool in Fig. (a).)

Because the sack of water is in static equilibrium, the magnitude $\vec{F}_b$ of is equal to the magnitude $m_f g$ of the gravitational force $\vec{F}_g$ on the sack of water: $F_b = m_f g$. (Subscript f refers to fluid, here the water.) In words, the magnitude of the buoyant force is equal to the weight of the water in the sack.

In Fig. (b), we have replaced the sack of water with a stone that exactly fills the hole in Fig. (a). The stone is said to displace the water, meaning that the stone occupies space that would otherwise be occupied by water. We have changed nothing about the shape of the hole, so the forces at the hole's surface must be the same as when the water-filled sack was in place. Thus, the same upward buoyant force that acted on the water-filled sack now acts on the stone; that is, the magnitude $\vec{F}_b$ of the buoyant force is equal to $m_f g$, the weight of the water displaced by the stone.

Unlike the water-filled sack, the stone is not in static equilibrium. The downward gravitational force $\vec{F}_g$ on the stone is greater in magnitude than the upward buoyant force, as is shown in the free-body diagram in Fig. (b). The stone thus accelerates downward, sinking to the bottom of the pool. Let us next exactly fill the hole in Fig. (a) with a block of lightweight wood, as in Fig. (c). Again, nothing has changed about the forces at the hole's surface, so the magnitude $\vec{F}_b$ of the buoyant force is still equal to $m_f g$, the weight of the displaced water. Like the stone, the block is not in static equilibrium. However, this time the gravitational force $\vec{F}_g$ is lesser in magnitude than the buoyant force (as shown to the right of the pool), and so the block accelerates upward, rising to the top surface of the water. Our results with the sack, stone, and block apply to all fluids and are summarized in **Archimedes' principle**:

When a body is fully or partially submerged in a fluid, a buoyant force $\vec{F}_b$ from the surrounding fluid acts on the body. The force is directed upward and has a magnitude equal to the weight $m_f g$ of the fluid that has been displaced by the body. The buoyant force on a body in a fluid has the magnitude

$$F_b = m_f g \text{ (buoyant force)}$$

where m_f is the mass of the fluid that is displaced by the body.

Floating

When we release a block of lightweight wood just above the water in a pool, the block moves into the water because the gravitational force on it pulls it downward.

As the block displaces more and more water, the magnitude F_b of the upward buoyant force acting on it increases. Eventually, F_b is large enough to equal the magnitude F_g of the downward gravitational force on the block, and the block comes to rest. The block is then in static equilibrium and is said to be floating in the water. In general, When a body floats in a fluid, the magnitude F_b of the buoyant force on the body is equal to the magnitude F_g of the gravitational force on the body.

We can write this statement as

$$F_b = F_g \text{ (Floating)}$$

From Eq. we know that $F_b = m_f g$. Thus,

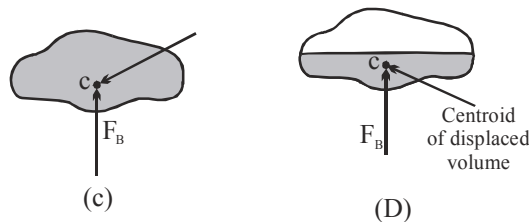
When a body floats in a fluid, the magnitude F_g of the gravitational force on the body is equal to the weight $m_f g$ of the fluid that has been displaced by the body.

We can write this statement as

$$F_g = m_f g$$

In other words, a floating body displaces its own weight of fluid.

The location of the line of action of the buoyant force can be determined by adding torques of the forces due to pressure forces, with respect to some convenient axis. The buoyant force must pass through the center of mass of the displaced volume, as shown in Fig. (c), as it was in translational and rotational equilibrium. The point through which the buoyant force acts is called the center of buoyancy.



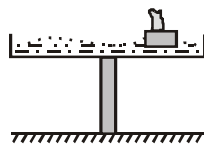
These same results apply to floating bodies which are only partially submerged, as shown in Fig.(d), if the density of the fluid above the liquid surface is very small compared with the liquid in which the body floats. Since the fluid above the surface is usually air, for practical purposes this condition is satisfied.

In the above discussion, the fluid is assumed to have a constant density. If a body is immersed in a fluid in which density varies with depth, such as having multiple layers of fluid, the magnitude of the buoyant force remains equal to the weight of the displaced fluid and the buoyant force passes through the center of mass of the displaced volume.

Q. A wooden block floats vertically in a glass filled with water. How will the level of the water in the glass change if the block is kept in a horizontal position ?

Solution : The level of the water will not change because the quantity of water displaced will remain the same.

Q. A vessel filled with water is placed exactly in middle of a thin wall (fig.). Will the system topple if a small wooden boat carrying some weight is floated in the vessel?



Solution : The system will not topple, since according to Pascal's law the pressure on the bottom of the vessel will be the same everywhere thus the body will still remain in rotational equilibrium

Ex. A homogeneous piece of ice floats in a glass filled with water. How will the level of the water in the glass change when the ice melts ?

Solution : Since the piece of ice floats, the weight of the water displaced by it is equal to the weight of the ice itself or the weight of the water it produces upon melting. For this reason the water formed by the piece of ice will occupy a volume equal to that of the submerged portion, and the level of the water will not change.

Q. A piece of ice is floating in a tub filled with water. How will the level of the water in the tub change when the ice melts ? Consider the following cases :

- (1) a stone is frozen in the ice
- (2) the ice contains an air bubble

- (1) The volume of the submerged portion of the piece with the stone is greater than the sum of the volumes of the stone and the water produced by the melting ice. Therefore, the level of the water in the glass will drop.
- (2) The weight of the displaced water is equal to that of the ice (the weight of the air in the bubble may be neglected). For this reason, as in conceptual eg., the level of the water will not change.

A vessel with a body floating in it is kept in elevator accelerating downwards with acceleration a such that $a < g$. Will the body rise or sink further in the vessel?

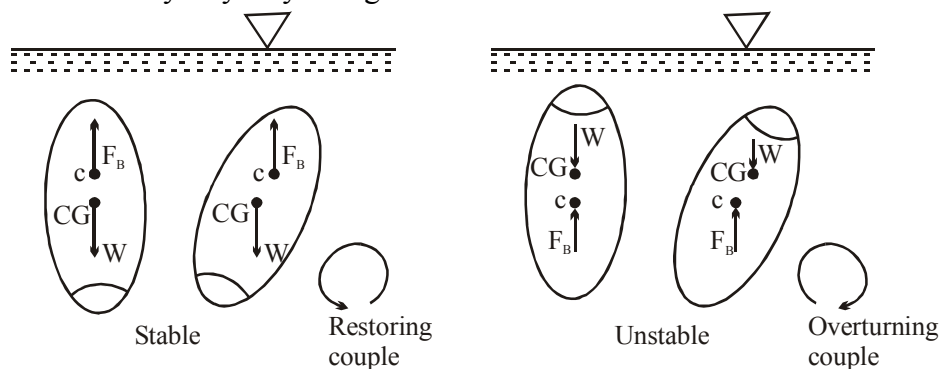
$$Mg - \rho V_0 (g - a) = Ma$$

Hence, $V_2 = \frac{M}{\rho}$ thus $V_2 = V$, as in a stationary vessel, $V = \frac{M}{\rho}$. Thus the body does not rise to the surface.

The center of buoyancy and center of gravity do not necessarily coincide so the floating or submerged body may not be in stable equilibrium. A small rotation can cause the buoyant force to produce either a restoring or overturning torque. For example, for the completely submerged body shown in Fig., which has a center of gravity below the center of buoyancy, a rotation from its equilibrium position will create a restoring torque by the buoyant force, F_B , which causes the body to rotate back to its original position. Thus, if the center of gravity falls below the center of buoyancy, the body is stable.

However, as shown in Fig., if the center of gravity of the completely submerged body is above the center of buoyancy, the resulting torque formed by the weight and the buoyant force will cause the body to overturn and move to a new equilibrium position. Thus, a completely submerged body with its center of gravity above its center of buoyancy is in an unstable equilibrium position.

For floating bodies the stability problem is more complicated, since as the body rotates the location of the center of buoyancy may change.



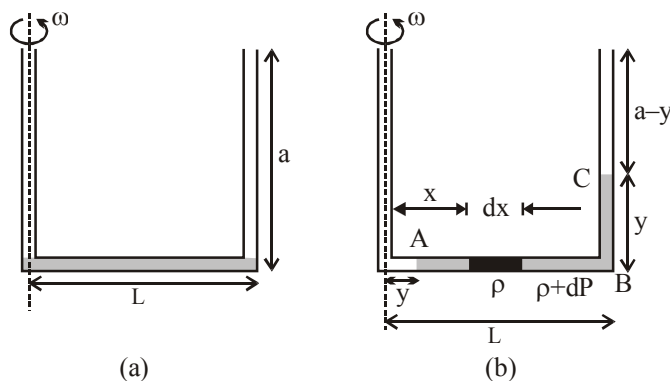
Concept : Length of a horizontal arm of a U-tube is L and ends of both the vertical arms are open to atmospheric pressure P_0 . A liquid of density ρ is poured in the tube such that liquid just fills the horizontal part of the tube as shown in figure. Now one end of the opened ends is sealed and the tube is then rotated about a vertical axis passing through the other vertical arm with angular speed ω . If length of each vertical arm is a and in the sealed end liquid rises to a height y , find pressure in the sealed tube during rotation.

The pressure difference across an element of width dx , which is given as

$$dP = dx\rho\omega^2x$$

Now integrating from A to B, we get

$$P_B - P_A = \int_y^L \rho\omega^2x dx$$



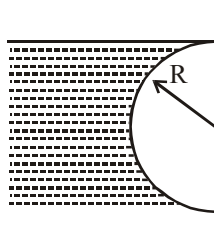
Thus pressure at point C can be given as

$$P_C = P_B - y\rho g$$

and at point A, pressure is atmospheric, thus we have

$$P_C = \frac{\rho\omega^2}{2}(L^2 - y^2) + P_A - y\rho g$$

- Q.** A hemisphere of radius R is just submerged is just sinking in water of density ρ . Find the
- horizontal thrust.
 - vertical thrust.
 - total hydrostatic force.
 - angle of orientation of total hydrostatic force acting on the hemisphere.
- Do not count atmospheric pressure

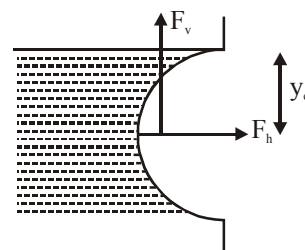


- Sol.** (a) Let the horizontal and vertical thrusts on the hemisphere be F_h and F_v respectively
We know that

$$F_h = \rho g y_c A_y$$

where

$$y_c = R$$



and $A_v = \pi R^2$

This gives $F_h = \rho g \pi R^3$ (right)

(b) Similarly using the formula $F_v = \rho V g$

Where $V =$ volume of the hemisphere $= \frac{2}{3} \pi R^3$,

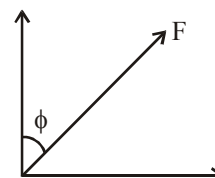
we have $F_v = \frac{2}{3} \rho g \pi R^3$ (up)

(c) Hence the net hydrostatic force on the hemisphere is

$$F = \sqrt{F_h^2 + F_v^2}$$

$$F = \sqrt{(\rho g \pi R^3)^2 + \left(\frac{2}{3} \rho g \pi R^3\right)^2}$$

$$= \frac{\sqrt{13}}{3} \rho g \pi R^3$$



(d) The angle of orientation of the force F is

$$\phi = \tan^{-1} \frac{F_h}{F_v}$$

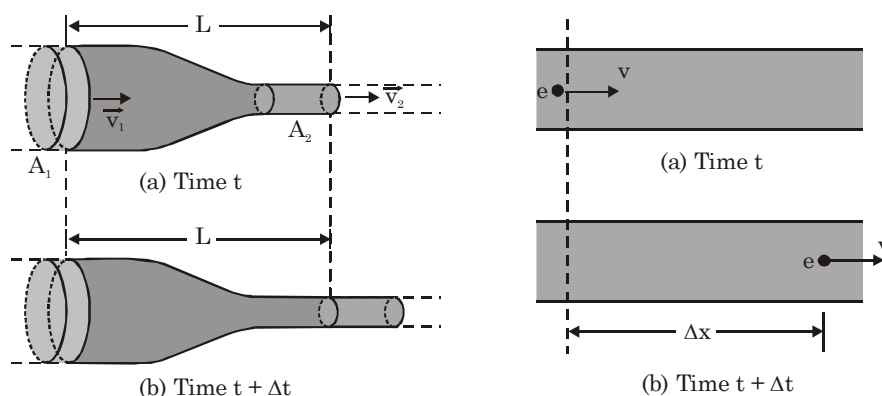
$$= \tan^{-1} \frac{\rho g \pi R^3}{\frac{2}{3} \rho g \pi R^3} = \tan^{-1} \frac{3}{2}$$

FLUID DYNAMICS & VISCOSITY

Ideal Fluids in Motion

The motion of real fluids is very complicated and not yet fully understood. Instead, we shall discuss the motion of an **ideal fluid**, which is simpler to handle mathematically and yet provides useful results. Here are four assumptions that we make about our ideal fluid; they all are concerned with flow:

1. **Steady flow** In steady (or laminar) flow, the velocity of the moving fluid at any fixed point does not change with time, either in magnitude or in direction. The gentle flow of water near the center of a quiet stream is steady; the flow in a chain of rapids is not. Figure shows a transition from steady flow to nonsteady (or nonlaminar or turbulent) flow for a rising stream of smoke. The speed of the smoke particles increases as they rise and, at a certain critical speed, the flow changes from steady to nonsteady.
2. **Incompressible flow** We assume, as for fluids at rest, that our ideal fluid is incompressible; that is, its density has a constant, uniform value.



$$\Delta V = A \Delta x = A v \Delta t.$$

Applying Eq. to both the left and right ends of the tube segment in Fig., we have

$$\Delta V = A_1 v_1 \Delta t = A_2 v_2 \Delta t$$

Or $A_1 v_1 = A_2 v_2$ (equation of continuity)

This relation between speed and cross-sectional area is called the **equation of continuity** for the flow of an ideal fluid. It tells us that the flow speed increases when we decrease the cross-sectional area through which the fluid flows (as when we partially close off a garden hose with a thumb). Equation applies not only to an actual tube but also to any so-called tube of flow, or imaginary tube whose boundary consists of streamlines. Such a tube acts like a real tube because no fluid element can cross a streamline; thus, all the fluid within a tube of flow must remain within its boundary. Figure shows a tube of flow in which the cross-sectional area increases from area A_1 to area A_2 along the flow direction. From Eq. we know that, with the increase in area, the speed must decrease, as is indicated by the greater spacing between streamlines at the right in Fig. . Similarly, you can see that in Fig. the speed of the flow is greatest just above and just below the cylinder. We can rewrite Eq. as

$R_v = A_v =$ a constant (volume flow rate, equation of continuity), in which R_v is the **volume flow rate** of the fluid (volume past a given point per unit time). Its SI unit is the cubic meter per second (m^3/s). If the density ρ of the fluid is uniform, we can multiply Eq. by that density to get the **mass flow rate** R_m (mass per unit time):

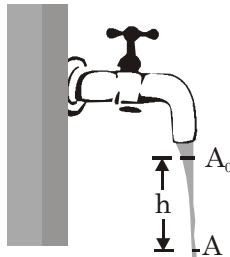
$$R_m = \rho R_v = A_v = \text{a constant (mass flow rate)}.$$

The SI unit of mass flow rate is the kilogram per second (kg/s). Equation says that the mass that flows into the tube segment of Fig. each second must be equal to the mass that flows out of that segment each second.

Sample Problem

Figure shows how the stream of water emerging from a faucet “necks down” as it falls. The indicated cross-sectional areas are $A_0 = 1.2 \text{ cm}^2$ and $A = 0.35 \text{ cm}^2$. The two levels are separated by a vertical distance $h = 45 \text{ mm}$. What is the volume flow rate from the tap?

The volume flow rate through the higher cross section must be the same as that through the lower cross section.



where v_0 and v are the water speeds at the levels corresponding to A_0 and A . From Eq. we can also write, because the water is falling freely with acceleration g ,

$$v^2 = v_0^2 + 2gh.$$

Eliminating v between Eqs. and solving for v_0 , we obtain

$$v_0 = \sqrt{\frac{2ghA^2}{A_0^2 - A^2}}$$

$$v_0 = \sqrt{\frac{(2) \times (9.8 \text{ m/s}^2)(0.045 \text{ m})(0.35 \text{ cm}^2)^2}{(1.2 \text{ cm}^2)^2 - (0.35 \text{ cm}^2)^2}}$$

$$v_0 = 0.286 \text{ m/s} = 28.6 \text{ cm/s}.$$

From Eq. , the volume flow rate R_V is then

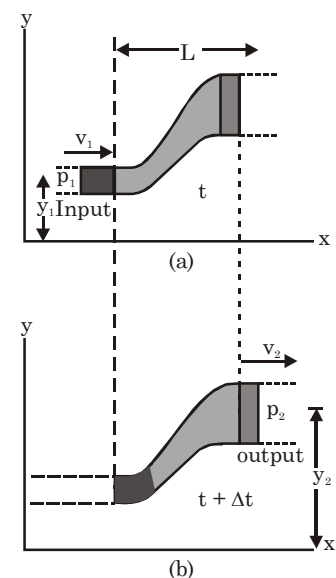
$$\begin{aligned} R_V &= A_0 v_0 = (1.2 \text{ cm}^2)(28.6 \text{ cm/s}) \\ &= 34 \text{ cm}^3/\text{s}. \text{ Ans. } \end{aligned}$$

Bernoulli's Equation

Figure represents a tube through which an ideal fluid is flowing at a steady rate. In a time interval Δt , suppose that a volume of fluid ΔV , colored purple in Fig. , enters the tube at the left (or input) end and an identical volume, colored green in Fig. , emerges at the right (or output) end. The emerging volume must be the same as the entering volume because the fluid is incompressible, with an assumed constant density ρ .

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

Let y_1 , v_1 , and p_1 be the elevation, speed, and pressure of the fluid entering at the left, and y_2 , v_2 , and p_2 be the corresponding quantities for the fluid emerging at the right. By applying the principle of conservation of energy to the fluid, we shall show that these quantities are related by



In general, the term $\frac{1}{2}\rho v^2$ is called the fluid's **kinetic energy**

density (kinetic energy per unit volume). We can also write Eq. as

$$p + \frac{1}{2}\rho v^2 + \rho gy \text{ a constant (Bernoulli's equation).}$$

Equations are equivalent forms of **Bernoulli's equation**, after Daniel Bernoulli, who studied fluid flow in the 1700s.* Like the equation of continuity Eq., Bernoulli's equation is not a new principle but simply the reformulation of a familiar principle in a form more suitable to fluid mechanics. As a check, let us apply Bernoulli's equation to fluids at rest, by putting $v_1 = v_2 = 0$ in Eq. The result is

$$p_2 = p_1 + \rho g(y_1 - y_2)$$

Which is equation.

A major prediction of Bernoulli's equation emerges if we take y to be a constant ($y = 0$, say) so that the fluid does not change elevation as it flows. Equation then becomes

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Which tells us that :

If the speed of a fluid element increases as the element travels along a horizontal streamline, the pressure of the fluid must decrease, and conversely.

Put another way, where the streamlines are relatively close together (where the velocity is relatively great), the pressure is relatively low, and conversely. The link between a change in speed and a change in pressure makes sense if you consider a fluid element. When the element nears a narrow region, the higher pressure behind it accelerates it so that it then has a greater speed in the narrow region. When it nears a wide region, the higher pressure ahead of it decelerates it so that it then has a lesser speed in the wide region. Bernoulli's equation is strictly valid only to the extent that the fluid is ideal. If viscous forces are present, thermal energy will be involved. We take no account of this in the derivation that follows.

Proof of Bernoulli's Equation

Let us take as our system the entire volume of the (ideal) fluid shown in Fig. . We shall apply the principle of conservation of energy to this system as it moves from its initial state (Fig. (a)) to its final state (Fig. (b)). The fluid lying between the two vertical planes separated by a distance L in Fig. does not change its properties during this process; we need be concerned only with changes that take place at the input and output ends. First, we apply energy conservation in the form of the work–kinetic energy theorem,

$$W = \Delta K,$$

which tells us that the change in the kinetic energy of our system must equal the net work done on the system. The change in kinetic energy results from the change in speed between the ends of the tube and is

$$\Delta K = \frac{1}{2}\Delta m v_2^2 - \frac{1}{2}\Delta m v_1^2 = \frac{1}{2}\rho\Delta V(v_2^2 - v_1^2)$$

in which $\Delta m (= \rho\Delta V)$ is the mass of the fluid that enters at the input end and leaves at the output end during a small time interval Δt .

The work done on the system arises from two sources. The work W_g done by the gravitational force ($\Delta m \vec{g}$) on the fluid of mass Δm during the vertical lift of the mass from the input level to the output level is

$$W_g = \Delta m g(y_2 - y_1) \\ = -\rho g \Delta V(y_2 - y_1)$$

This work is negative because the upward displacement and the downward gravitational force have opposite directions.

Work must also be done on the system (at the input end) to push the entering fluid into the tube and by the system (at the output end) to push forward the fluid that is located ahead of the emerging fluid. In general, the work done by a force of magnitude F , acting on a fluid sample contained in a tube of area A to move the fluid through a distance Δx , is

$$F \Delta x = (pA)(\Delta x) = p(A \Delta x) = p \Delta V$$

The work done on the system is then $p_1 \Delta V$, and the work done by the system is $-p_2 \Delta V$. Their sum W_p is W_p

$$W_p = -p_2 \Delta V - p_1 \Delta V \\ = -(p_2 - p_1) \Delta V.$$

The work-kinetic energy theorem of Eq. now becomes

$$W = W_g + W_p = \Delta K.$$

Substituting from Eqs. yields

$$-\rho g \Delta V(y_2 - y_1) - \Delta V(p_2 - p_1) = \frac{1}{2} \rho \Delta V(v_2^2 - v_1^2)$$

This, after a slight rearrangement, matches Eq. , which we set out to prove.

Sample Problem : In the old West, a desperado fires a bullet into an open water tank (Fig.), creating a hole a distance h below the water surface. What is the speed v of the water exiting the tank?

Sol. From Eq. $R_v = av = Av_0$ and thus

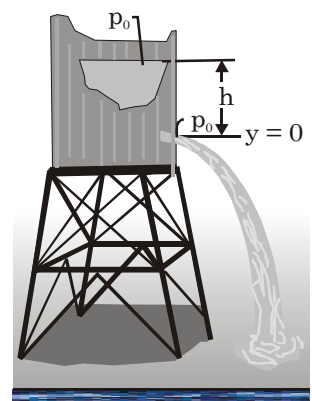
$$v_0 = \frac{a}{A} v$$

Because $a \ll A$, we see that $v_0 \ll v$. To apply Bernoulli's equation, we take the level of the hole as our reference level for measuring elevations (and thus gravitational potential energy). Noting that the pressure at the top of the tank and at the bullet hole is the atmospheric pressure p_0 (because both places are exposed to the atmosphere), we write Eq. as

$$p_0 + \frac{1}{2} \rho v_0^2 + \rho gh = p_0 + \frac{1}{2} \rho v^2 + \rho g(0)$$

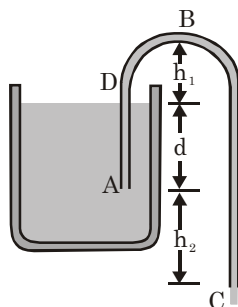
(Here the top of the tank is represented by the left side of the equation and the hole by the right side. The zero on the right indicates that the hole is at our reference level.) Before we solve Eq. for v , we can use our result that $v_0 \ll v$ to simplify it: We assume that v_0^2 and thus the term $\frac{1}{2} \rho v_0^2$ in Eq. , is negligible relative to the other terms, and we drop it. Solving the remaining equation for v then yields

$$v = \sqrt{2gh} \text{ Ans.}$$



This is the same speed that an object would have when falling a height h from rest. This is because the work done by atmospheric pressure is cancelling out at open surface and the hole.

Sample Problem : shows a siphon, which is a device for removing liquid from a container. Tube ABC must initially be filled, but once this has been done, liquid will flow through the tube until the liquid surface in the container is level with the tube opening at A. The liquid has density 1000 kg/m^3 and negligible viscosity. The distances shown are $h_1 = 25 \text{ cm}$, $d = 12 \text{ cm}$, and $h_2 = 40 \text{ cm}$. (a) With what speed does the liquid emerge from the tube at C? (b) If the atmospheric pressure is $1.0 \times 10^5 \text{ Pa}$, what is the pressure in the liquid at the topmost point B? (c) Theoretically, what is the greatest possible height h_1 that a siphon can lift water?



Solution: You may have used siphon and you may recollect that lower the exit point of the fluid is, faster the fluid flows out.

We consider a point D on the surface of the liquid in the container, in the same tube of flow with points A, B and C. Applying Bernoulli's equation to points D and C, we obtain

$$p_D + \frac{1}{2} \rho v_D^2 + \rho g h_D = p_C + \frac{1}{2} \rho v_C^2 + \rho g h_C$$

$$v_C = \sqrt{\frac{2(p_D - p_C)}{\rho} + 2g(h_D - h_C) + v_D^2}$$

$$\approx \sqrt{2g(d + h_2)}$$

where in the last step we set $p_D = p_C = p_{\text{air}}$ and $v_D/v_C \approx 0$. Plugging in the values, we obtain

$$v_c = \sqrt{2(9.8 \text{ m/s}^2)(0.40 \text{ m} + 0.12 \text{ m})} = 3.2 \text{ m/s}$$

The result confirms our experience.

We now consider points B and C:

$$p_B + \frac{1}{2} \rho v_B^2 + \rho g h_B = p_C + \frac{1}{2} \rho v_C^2 + \rho g h_C$$

Since $v_B = v_C$ by equation of continuity, and $p_C = p_{\text{air}}$, Bernoulli's equation becomes

$$p_B = p_C + \rho g(h_C - h_B) = p_{\text{air}} - \rho g(h_1 + h_2 + d)$$

$$= 1.0 \times 10^5 \text{ Pa} - (1.0 \times 10^3 \text{ kg/m}^3)(9.8 \text{ m/s}^2)(0.25 \text{ m} + 0.40 \text{ m} + 0.12 \text{ m})$$

$$= 9.2 \times 10^4 \text{ Pa.}$$

Since $p_B \geq 0$, we must let $p_{\text{air}} - \rho g(h_1 + d + h_2) \geq 0$, which yields

$$h_1 \leq h_{1\text{max}} = \frac{p_{\text{air}}}{\rho} - d - h_2 \leq \frac{p_{\text{air}}}{\rho} = 10.3 \text{ m}$$

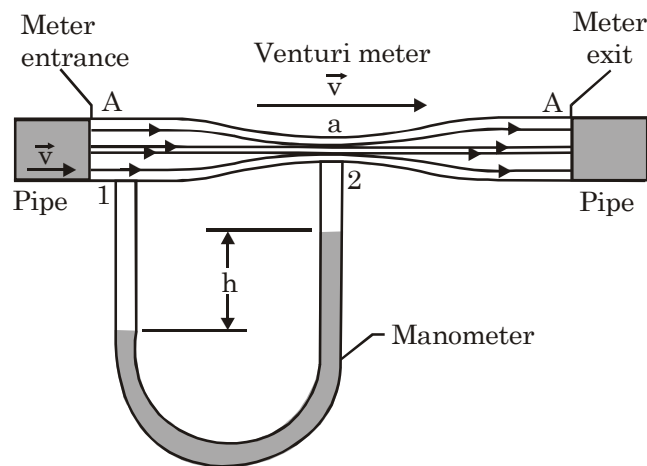
or
$$v_f = \sqrt{\frac{2\rho gh}{\sigma}}$$

Thus we can see that we have measured the fluid velocity as this was the only difference between the two tubes leading to the pressure difference between the tubes.

Conceptual example: A venturi meter is used to measure the flow speed of a fluid in a pipe. The meter is connected between two sections of the pipe (Fig.); the cross-sectional area A of the entrance and exit of the meter matches the pipe's cross-sectional area. Between the entrance and exit, the fluid flows from the pipe with speed V and then through a narrow "throat" of cross-sectional area a with speed v . A manometer connects the wider portion of the meter to the narrower portion. The change in the fluid's speed is accompanied by a change Δp in the fluid's pressure, which causes a height difference h of the liquid in the two arms of the manometer. (Here Δp means pressure in the throat minus pressure in the pipe.) (a) By applying Bernoulli's equation and the equation of continuity to points 1 and 2 in Fig. , show that

$$v = \sqrt{\frac{2a^2 \Delta p}{\rho(a^2 - A^2)}}$$

where ρ is the density of the fluid. (b) Suppose that the fluid is fresh water, that the cross-sectional areas are 64 cm^2 in the pipe and 32 cm^2 in the throat, and that the pressure is 55 kPa in the pipe and 41 kPa in the throat. What is the rate of water flow in cubic meters per second?



- (a) The continuity equation yields $Av = aV$, and Bernoulli's equation yields $\Delta p + \frac{1}{2}\rho v^2 = \frac{1}{2}\rho V^2$ where $\Delta p = p_1 - p_2$. The first equation gives $V = (A/a)v$. We use this to substitute for V in the second equation, and obtain $\Delta p + \frac{1}{2}\rho v^2 = \frac{1}{2}\rho(A/a)^2 v^2$. We solve for v . The result is

$$v = \sqrt{\frac{2\Delta p}{\rho((A/a)^2 - 1)}} = \sqrt{\frac{2a^2 \Delta p}{\rho(A^2 - a^2)}}$$

- (b) We substitute values to obtain

$$v = \sqrt{\frac{2(32 \times 10^{-4} \text{ m}^2)^2 (55 \times 10^3 \text{ Pa} - 41 \times 10^3 \text{ Pa})}{(1000 \text{ kg/m}^3)((64 \times 10^{-4} \text{ m}^2)^2 - (32 \times 10^{-4} \text{ m}^2)^2)}} = 30.06 \text{ m/s}$$

Consequently, the flow rate is

$$Av = (64 \times 10^{-4} \text{ m}^2) (3.06 \text{ m/s}) = 2.0 \times 10^{-2} \text{ m}^3 / \text{s}.$$

SURFACE TENSION & VISCOSITY

EXPLANATION OF SOME OBSERVED PHENOMENA

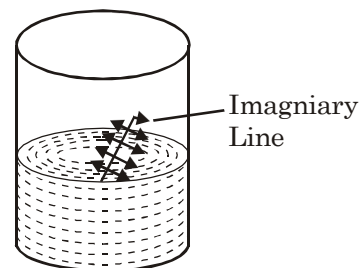
1. Lead balls are spherical in shape.
2. Rain drops and a globule of mercury placed on glass plate are spherical.
3. Hair of a shaving brush/painting brush, when dipped in water spread out, but as soon as it is taken out. Its hair stick together.
4. A greased needle placed gently on the free surface of water in a beaker does not sink.
5. Similarly, insects can walk on the free surface of water without drowning.
6. Bits of Camphor gum move irregularly when placed on water surface.

SURFACE TENSION

Surface Tension is a property of liquid at rest by virtue of which a liquid surface gets contracted to a minimum area and behaves like a stretched membrane.

Surface Tension of a liquid is measured by force per unit length on either side of any imaginary line drawn tangentially over the liquid surface, force being normal to the imaginary line as shown in fig. i.e. Surface tension.

$$(T) = \frac{\text{Total force on either of the imaginary line (F)}}{\text{Length of the line (l)}}$$



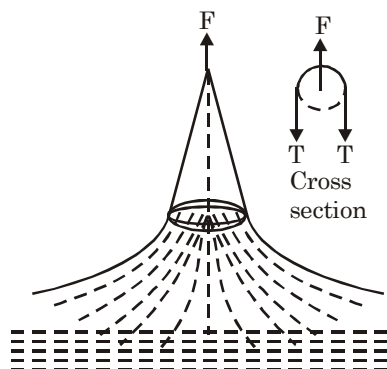
Unit of Surface Tension

In C. G. S. system the unit of surface tension is dyne/cm (dyne cm⁻¹) and SI system its units is Nm⁻¹

Solved Examples

- Ex.1 A ring is cut from a platinum tube of 8.5 cm internal and 8.7 cm external diameter. It is supported horizontally from a pan of a balance so that it comes in contact with the water in a glass vessel. What is the surface tension of water if an extra 3.97 g weight is required to pull it away from water? ($g = 980 \text{ cm/s}^2$).

Sol.



The ring is in contact with water along its inner and outer circumference ; so when pulled out the total force on it due to surface tension will be

$$F = T(2\pi r_1 + 2\pi r_2)$$

$$\text{So, } T = \frac{mg}{2\pi(r_1 + r_2)} \quad [\because F = mg]$$

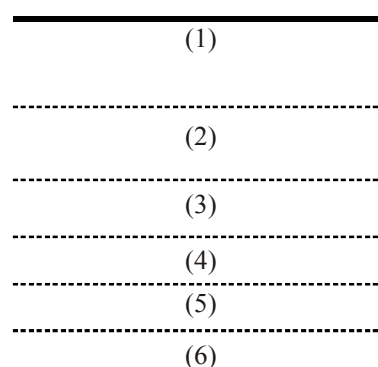
$$\text{i.e., } T = \frac{3.97 \times 980}{3.14 \times (8.5 + 8.7)} = 72.13 \text{ dyne/cm}$$

Surface energy

The course of reasoning given below is usually followed to prove that the molecules of the surface layer of a liquid have surplus potential energy. A molecule inside the liquid is acted upon by the forces of attraction from the other molecules which compensate each other on the average. If a molecule is singled out on the surface, the resulting force of attraction from the other molecule is directed into the liquid. For this reason the molecule tends to move into the liquid, and definite work should be done to bring it to the surface. Therefore, each molecule of the surface layer has excess potential energy equal to this work. The average force that acts on any molecule from the side of all the others, however, is always equal to zero if the liquid is in equilibrium. This is why the work done to move the liquid from a depth to the surface should also be zero. What is the origin, in this case, of the surface energy?

[Sol. The forces of attraction acting on a molecule in the surface layer from all the other molecules produce a resultant directed downward. The closest neighbours, however, exert a force of repulsion on the molecule which is therefore in equilibrium.

Owing to the forces of attraction and repulsion, the density of the liquid is smaller in the surface layer than inside. Indeed, molecule 1 (figure) is acted upon by the force of repulsion from molecule 2 and the forces of attraction from all the other molecules (3, 4,). Molecule 2 is acted upon by the forces of repulsion from 3 and 1 and the forces of attraction from the molecules in the deep layers. As a result, distance 1-2 should be greater than 2-3, etc.



This course of reasoning is quite approximate (thermal motion, etc. is disregarded), but nevertheless it gives a qualitatively correct result.

An increase in the surface of the liquid causes new sections of the rarefied surface layer to appear. Here work should be performed against the forces of attraction between the molecules. It is this work that constitutes the surface energy.

We know that the molecules on the liquid surface experience net downward force. So to bring a molecule from the interior of the liquid to the free surface, some work is required to be done against the intermolecular force of attraction, which will be stored as potential energy of the molecule on the surface. The potential energy of surface molecules per unit area of the surface is called surface energy. Unit of surface energy is erg cm^{-2} in C.G.S. system and Jm^{-2} in SI system. Dimensional formula of surface energy is $[\text{ML}^0\text{T}^{-2}]$ surface energy depends on number of surfaces e.g. a liquid drop is having one liquid air surface while bubble is having two liquid air surface.

Relation between surface tension and surface energy

Consider a rectangular frame PQRS of wire, whose arm RS can slide on the arms PR and QS. If this frame is dipped in a soap solution, then a soap film is produced in the frame PQRS in fig. Due to surface tension (T), the film exerts a force on the frame (towards the interior of the film). Let l be the length of the arm RS, then the force acting on the arm RS towards the film is $F = T \times 2l$ [Since soap film has two surfaces, that is why the length is taken twice.]

$\therefore$ work done, $W = Fx = 2Tlx$

Increase in potential energy of the soap film.

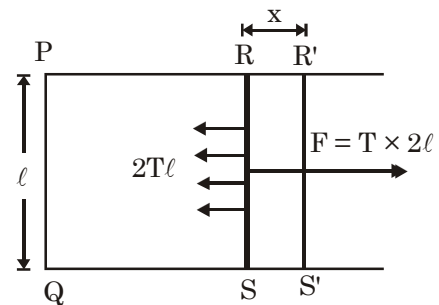
$= EA = 2Elx = \text{work done in increasing the area } (\Delta W)$

where E = surface energy of the soap film per unit area.

According to the law of conservation of energy, the work done must be equal to the increase in the potential energy.

$$\therefore 2Tlx = 2Elx \text{ or } T = E = \frac{\Delta W}{\Delta A}$$

Thus, surface tension is numerically equal to surface energy or work done per unit increase surface area.



Ex.11 A mercury drop of radius 1 cm is sprayed into 10^6 droplets of equal size. Calculate the energy expanded if surface tension of mercury is 35×10^{-3} N/m.

Sol. If drop of radius R is sprayed into n droplets of equal radius r , then as a drop has only one surface, the initial surface area will be $4\pi R^2$ while final area is $n(4\pi r^2)$. So the increase in area

$$\Delta S = n(4\pi r^2) - 4\pi R^2$$

So energy expended in the process.

$$W = T\Delta S = 4\pi T [nr^2 - R^2] \quad \dots (1)$$

Now since the total volume of n droplets is the same as that of initial drop, i.e.

$$\frac{4}{3} \pi R^3 = n \left[\frac{4}{3} \pi r^3 \right] \text{ or } r = R/n^{1/3} \quad \dots (2)$$

Putting the value of r from equation (2) in (1)

$$W = 4\pi R^2 T (n^{1/3} - 1)$$

Ex.12 If a number of little droplets of water, each of radius r , coalesce to form a single drop of radius R , show that the rise in temperature will be given by

$$\frac{3T}{J} \left(\frac{1}{r} - \frac{1}{R} \right)$$

where T is the surface tension of water and J is the mechanical equivalent of heat.

Sol. Let n be the number of little droplets.

Since volume will remain constant, hence volume of n little droplets = volume of single drop

$$\therefore n \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3 \text{ or } nr^3 = R^3$$

Decrease in surface area = $n \times 4\pi r^2 - 4\pi R^2$

$$\text{or } \Delta A = 4\pi [nr^2 - R^2] = 4\pi \left[\frac{nr^3}{r} - R^2 \right] = 4\pi \left[\frac{R^3}{r} - R^2 \right] = 4\pi R^3 \left[\frac{1}{r} - \frac{1}{R} \right]$$

$$\text{Energy evolved } W = T \times \text{decrease in surface area} = T \times 4\pi R^3 \left[\frac{1}{r} - \frac{1}{R} \right]$$

$$\text{Heat produced, } Q = \frac{W}{J} = \frac{4\pi TR^3}{J} \left[\frac{1}{r} - \frac{1}{R} \right] \quad \text{But } Q = ms d\theta$$

where m is the mass of big drop, s is the specific heat of water and $d\theta$ is the rise in temperature.

$$\therefore \frac{4\pi TR^3}{J} \left[\frac{1}{r} - \frac{1}{R} \right] = \text{volume of big drop} \times \text{density of water} \times \text{sp. heat of water} \times d\theta$$

$$\text{or, } \frac{4}{3} \pi R^3 \times 1 \times 1 \times d\theta = \frac{4\pi TR^3}{J} \left(\frac{1}{r} - \frac{1}{R} \right) \quad \text{or, } d\theta = \frac{3T}{J} \left[\frac{1}{r} - \frac{1}{R} \right]$$

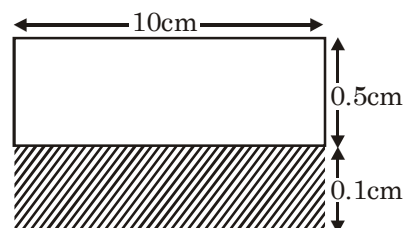
Ex. 13 A film of water is formed between two straight parallel wires each 10cm long and at a separation 0.5 cm. Calculate the work required to increase 1mm distance between them. Surface tension of water $72 \times 10^{-3} \text{ N/m}$.

Sol. Here the increase in area is shown by shaded portion in the figure.

Since this a water film, it has two surface, therefore increase in area, $\Delta S = 2 \times 10 \times 0.1 = 2\text{cm}^2$

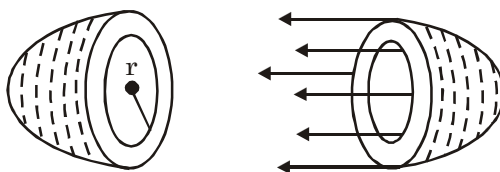
$\therefore$ Work required to be done

$$\begin{aligned} W &= \Delta S \times T \\ &= 2 \times 10^{-4} \times 72 \times 10^{-3} \\ &= 144 \times 10^{-7} \text{ joule} \\ &= 1.44 \times 10^{-5} \text{ joule} \end{aligned}$$



Excess pressure inside A liquid drop and a bubble

- Inside a bubble : Consider a soap bubble of radius r . Let p be the pressure inside the bubble and p_a outside. The excess pressure $= p - p_a$. Imagine the bubble broken into two halves, and consider one half of it as shown in fig. Since there are two surface, inner and outer, so the force due to surface tension is



$$F = \text{surface tension} \times \text{length} = T \times 2 (\text{circumference of the bubble}) = T \times 2 (2\pi r) \quad \dots (1)$$

The excess pressure $(p - p_a)$ acts on a cross-sectional area πr^2 , so the force due to excess pressure is

$$\Rightarrow F = (p - p_a) \pi r^2 \quad \dots (2)$$

The surface tension force given by equation (1) must balance the force due to excess pressure given by equation (2) to maintain the equilibrium, i.e. $(p - p_a) \pi r^2 = T \times 2(2\pi r)$

$$\text{or} \quad (p - p_a) = \frac{4T}{r} = p_{\text{excess}}$$

above expression can also be obtained by equation of excess pressure of curve surface by putting $R_1 = R_2$.

- Inside the drop : In a drop, there is only one surface and hence excess pressure can be written as

$$(p - p_a) = \frac{2T}{r} = p_{\text{excess}}$$

$$\text{or, } \left[(P + h_1 \rho g) + \frac{2T}{r_1} \right] r_1^3 = \left[(P + h_2 \rho g) + \frac{2T}{r_2} \right] r_2^3$$

Given that : $h_1 = 100$ cm, $r_1 = 0.1$ mm = 0.01 cm, $r_2 = 0.126$ cm, $T = 567$ dyne / cm, $P = 76$ cm of mercury. Substituting all the values, we get

$$h_2 = 9.48 \text{ cm.}$$

THE FORCE OF COHESION

The force of attraction between the molecules of the same substance is called cohesion.

In case of solids, the force of cohesion is very large and due to this solids have definite shape and size. On the other hand, the force of cohesion in case of liquids is weaker than that of solids. Hence liquids do not have definite shape but have definite volume. The force of cohesion is negligible in case of gases. Because of this fact, gases have neither fixed shape nor volume.

Example

- (i) Two drops of a liquid coalesce into one when brought in mutual contact because of the cohesive force.
- (ii) It is difficult to separate two sticky plates of glass wetted with water because a large force has to be applied against the cohesive force between the molecules of water.
- (iii) It is very difficult to break a drop of mercury into small droplets because of large cohesive force between mercury molecules.

FORCE OF ADHESION

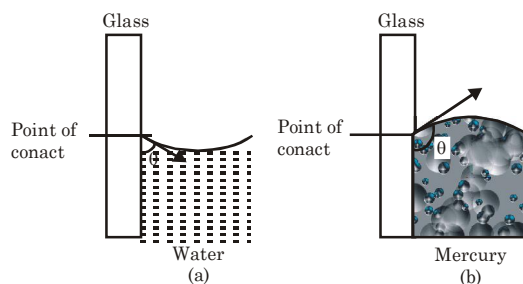
The force of attraction between molecules of different substance is called adhesion.

Example

- (i) Adhesive force enables us to write on the black board with a chalk.
- (ii) Adhesive force helps us to write on the paper with ink.
- (iii) Large force of adhesion between cement and bricks helps us in construction work.
- (iv) Due to force of adhesive, water wets the glass plate.
- (v) Fevicol and gum are used in gluing two surfaces together because of adhesive force.

ANGLE OF CONTACT

The angle which the tangent to the liquid surface at the point of contact makes with the solid surface inside the liquid is called angle of contact. Those liquids which wet the wall of the container (say in case of water and glass) have meniscus concave upwards and their value of angle of contact is less than 90° (also called acute angle). However, those liquids which don't wet the walls of the container (say in case of mercury and glass) have meniscus convex upwards and their value of angle of contact is greater than 90° (also called obtuse angle). The angle of contact of mercury with glass about 140° , whereas the angle of contact of water with glass is about 8° . But, for pure water, the angle of contact θ with glass is taken as 0° .



Solved Examples

Ex.3 A barometer contains two uniform capillaries of radii 1.44×10^{-3} m and 7.2×10^{-4} m. if the height of the liquid in the tube is 0.2m more than that in the wide tube, calculate the true pressure difference. Density of liquid = 10^3 kg/m³, surface tension = 72×10^{-3} N/m and $g = 9.8$ m/s².

Sol. Let the pressure in the wide and narrow capillaries of radii r_1 and r_2 respectively be P_1 and P_2 . Then pressure just below the meniscus in the wide and narrow tubes respectively are :

$$\left(P_1 - \frac{2T}{r_1} \right) \text{ and } \left(P_2 - \frac{2T}{r_2} \right) \quad \left[\text{excess pressure} = \frac{2T}{r} \right]$$

$$\text{Difference in these pressure} = \left(P_1 - \frac{2T}{r_1} \right) - \left(P_2 - \frac{2T}{r_2} \right) = h\rho g$$

$$\therefore \quad \text{True pressure difference} = P_1 - P_2$$

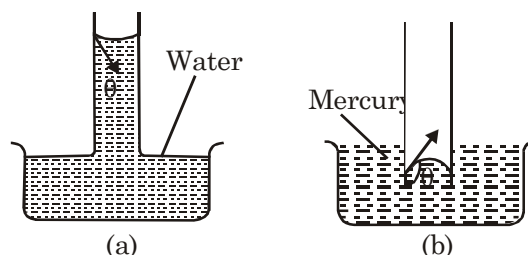
$$= h\rho g + 2T \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$= 0.2 \times 10^3 \times 9.8 + 2 \times 72 \times 10^{-3} \left[\frac{1}{1.44 \times 10^{-3}} - \frac{1}{7.2 \times 10^{-4}} \right]$$

$$= 1.86 \times 10^3 = 1860 \text{ N/m}^2$$

Capillarity

A glass tube of very fine bore throughout the length of the tube is called capillary tube. If the capillary tube is dipped in water, the water wets the inner side of the tube and rises in it [shown in figure (a)]. If the same capillary tube is dipped in the mercury, then the mercury is depressed [shown in figure (b)]. The phenomenon of rise or fall of liquids in a capillary tube is called capillarity.



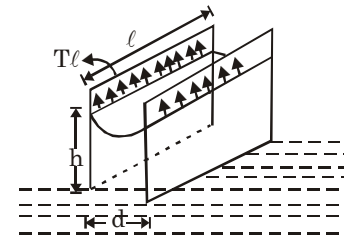
Particle applications of capillarity

1. The oil in a lamp rises in the wick by capillary action.
2. The tip of nib of a pen is split up, to make a narrow capillary so that the ink rises upto the tin or nib continuously.
3. Sap and water rise upto the top of the leaves of the tree by capillary action.
4. If one end of the towel dips into a bucket of water and the Other end hangs over the bucket the towel soon becomes wet throughout due to capillary action.
5. Ink is absorbed by the blotter due to capillary action.
6. Sandy soil is more dry than clay. It is because the capillaries between sand particles are not so fine as to draw the water up by capillaries.
7. The moisture rises in the capillaries of soil to the surface, where it evaporates. To preserve the moisture in the soil, capillaries must be broken up. This is done by ploughing and leveling the fields.
8. Bricks are porous and behave like capillaries.

- (iv) If two parallel plates with the spacing 'd' are placed in water reservoir, then height of rise

$$\Rightarrow 3Tl = \rho l h d g$$

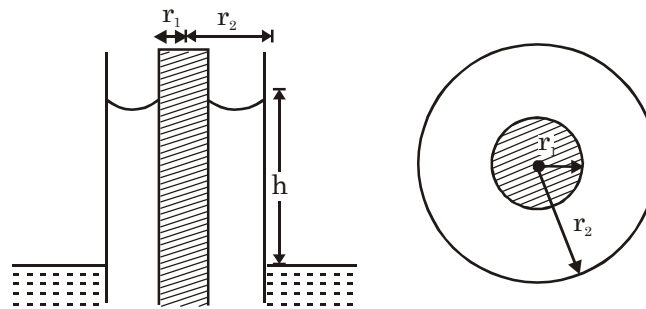
$$h = \frac{2T}{\rho d g}$$



- (v) If two concentric tube of radius 'r₁' and 'r₂' (inner one is solid) are placed in water reservoir, then height of rise

$$\Rightarrow T [2\pi r_1 + 2\pi r_2] = [\pi r_2^2 h - \pi r_1^2 h] \rho g$$

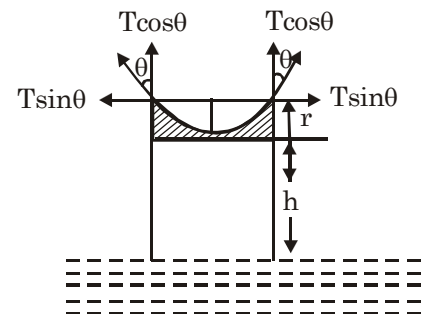
$$h = \frac{2T}{(r_2 - r_1)\rho g}$$



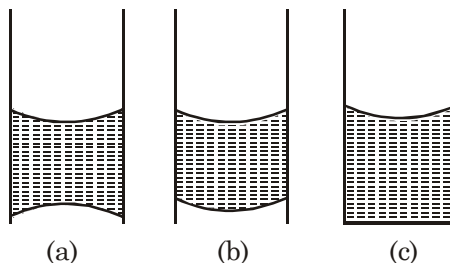
- (vi) If weight of the liquid in the meniscus is to be consider :

$$T \cos \theta \times 2\pi r = [\pi r^2 h + \frac{1}{3} \pi r^2 \times \pi r_1^2 h] \rho g$$

$$\left[h + \frac{r}{3} \right] = \frac{2T \cos \theta}{r \rho g}$$



- (vii) When capillary tube (radius, 'r') is in vertical position, the upper meniscus is concave and pressure due to surface tension is directed vertically upward and is given by $p_1 = 2T / R_1$ where R_1 = radius of curvature of upper meniscus.



The hydrostatic pressure $p_2 = h\rho g$ is always directed downwards.

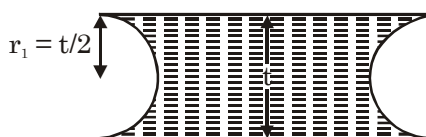
If $p_1 > p_2$ i.e. resulting pressure is directed upward. For equilibrium, the pressure due to lower meniscus should be downward. This makes lower meniscus concave downward (fig. (a)). The

radius of lower meniscus R_2 can be given by $\frac{2T}{R_2} = (p_1 - p_2)$.

Solved Examples

Ex.5 A drop of water volume 0.05 cm^3 is pressed between two glass-plates, as a consequence of which, it spreads and occupies an area of 40 cm^2 . If the surface tension of water is 70 dyne/cm , find the normal force required to separate out the two glass plates in newton.

Sol. Pressure inside the film is less than outside by an amount, $P = T \left[\frac{1}{r_1} + \frac{1}{r_2} \right]$, where r_1 and r_2 are the radii of curvature of the meniscus. Here $r_1 = t/2$ and $r_2 = \infty$, then the force required to separate the two glass plates, between which a liquid film is enclosed (figure) is, $F = P \times A = \frac{2AT}{t}$, where t is the thickness of the film, A = area of film.



$$F = \frac{2A^2T}{At} = \frac{2A^2T}{V} = \frac{2 \times (40 \times 10^{-4})^2 \times (70 \times 10^{-3})}{0.05 \times 10^{-6}} = 45 \text{ N}$$

Ex.6 A glass plate of length 10 cm , breadth 1.54 cm and thickness 0.20 cm weigh 8.2 gm in air. It is held vertically with the long side horizontal and the lower half under water. Find the apparent weight of the plate. Surface tension of water 73 dyne per cm , $g = 980 \text{ cm/sec}^2$.

Sol. Volume of the portion of the plate immersed in water is

$$10 \times \frac{1}{2} (1.54) \times 0.2 = 1.54 \text{ cm}^3.$$

Therefore, if the density of water is taken as 1, then upthrust

= wt. of the water displaced

$$= 1.54 \times 1 \times 980 = 1509.2 \text{ dynes.}$$

Now, the total length of the plate in contact with the water surface is $2 (10 + 0.2) = 20.4 \text{ cm}$,

∴ downward pull upon the plate due to surface tension

$$= 20.4 \times 73 = 1489.2 \text{ dynes}$$

∴ resultant upthrust

$$= 1509.2 - 1489.2$$

$$= 20.0 \text{ dynes} = \frac{20}{980}$$

$$= 0.0204 \text{ gm. wt.}$$

∴ apparent weight of the plate in water

$$= \text{weight of the plate in air} - \text{resultant upthrust}$$

$$= 8.2 - 0.0204 = 8.1796 \text{ gm Ans.}$$

Ex.7 A glass tube of circular cross-section is closed at one end. This end is weighted and the tube floats vertically in water, heavy end down. How far below the water surface is the end of the tube? Give : Outer radius of the tube 0.14 cm, mass of weighted tube 0.2 gm, surface tension of water 73 dyne/cm and $g = 980 \text{ cm/sec}^2$.

Sol., Let l be the length of the tube inside water. The forces acting on the tube are :

(i) Upthrust of water acting upward

$$= \pi r^2 l \times 1 \times 980 = \frac{22}{7} \times (0.14)^2 l \times 980 = 60.368 l \text{ dyne.}$$

(ii) Weight of the system acting downward

$$= mg = 0.2 \times 980 = 196 \text{ dyne.}$$

(iii) Force of surface tension acting downward

$$= 2\pi rT$$

$$= 2 \times \frac{22}{7} \times 0.14 \times 73 = 64.24 \text{ dyne.}$$

Since the tube is in equilibrium, the upward force is balanced by the downward forces. That is,

$$60.368 l = 196 + 64.24 = 260.24.$$

$$\therefore l = \frac{260.24}{60.368} = 4.31 \text{ cm.}$$

Ex.8 A glass U-tube is such that the diameter of one limb is 3.0 mm and that of the other is 6.00 mm. The tube is inverted vertically with the open ends below the surface of water in a beaker. What is the difference between the heights to which water rises in the two limbs? Surface tension of water is 0.07 Nm^{-1} . Assume that the angle of contact between water and glass is 0° .

Sol. Suppose pressure at the points A, B, C and D be P_A , P_B , P_C and P_D respectively.

The pressure on the concave side of the liquid surface is greater than that on the other side by $2T/R$.

Angle of contact θ is given to be 0° , hence $R \cos 0^\circ = r$ or $R = r$

$$\therefore P_A = P_B + 2T/r_1 \text{ and } P_C = P_D + 2T/r_2$$

where r_1 and r_2 are the radii of the two limbs

$$\text{But } P_A = P_C$$

$$\therefore P_B + \frac{2T}{r_1} = P_D + \frac{2T}{r_2}$$

$$\text{or } P_D - P_B = 2T \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

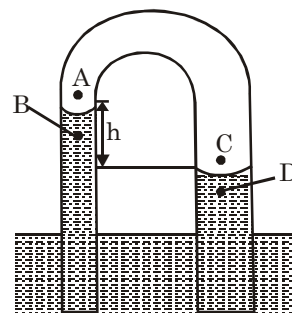
where h is the difference in water levels in the two limbs

$$\text{Now, } h = \frac{2T}{\rho g} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

Given that $T = 0.07 \text{ Nm}^{-1}$, $\rho = 1000 \text{ kgm}^{-3}$

$$r_1 = \frac{3}{2} \text{ mm} = \frac{3}{20} \text{ cm} = \frac{3}{20 \times 100} \text{ m} = 1.5 \times 10^{-3} \text{ m}, r_2 = 3 \times 10^{-3} \text{ m}$$

$$\therefore h = \frac{2 \times 0.07}{1000 \times 9.8} \left(\frac{1}{1.5 \times 10^{-3}} - \frac{1}{3 \times 10^{-3}} \right) \text{ m} = 4.76 \times 10^{-3} \text{ m} = 4.76 \text{ mm}$$



Ex.9 Two narrow bores of diameters 3.0 mm and 6.0 mm are joined together to form a U-shaped tube open at both ends. If the U-tube contains water, what is the difference in its levels in the two limbs of the tube? Surface tension of water at the temperature of the experiment is $7.3 \times 10^{-2} \text{ Nm}^{-1}$. Take the angle of contact to be zero.

Sol. Given that $r_1 = \frac{3.0}{2} = 1.5 \text{ mm} = 1.5 \times 10^{-3} \text{ m}$, $r_2 = \frac{6.0}{2} = 3.0 \times 10^{-3} \text{ m}$,

$$T = 7.3 \times 10^{-2} \text{ Nm}^{-1}, \theta = 0^\circ, \rho = 1.0 \times 10^3 \text{ kg m}^{-3}, g = 9.8 \text{ ms}^{-2}$$

When angle of contact is zero degree, the radius of the meniscus equals radius of bore.

$$\text{Excess pressure in the first bore, } P_1 = \frac{2T}{r_1} = \frac{2 \times 7.3 \times 10^{-2}}{1.5 \times 10^{-3}} = 97.3 \text{ Pascal}$$

$$\text{Excess pressure in the second bore, } P_2 = \frac{2T}{r_2} = \frac{2 \times 7.3 \times 10^{-2}}{3 \times 10^{-3}} = 48.7 \text{ Pascal}$$

Hence, pressure difference in the two limbs of the tube

$$\Delta P = P_1 - P_2 = h\rho g$$

$$\text{or } h = \frac{P_1 - P_2}{\rho g} = \frac{97.3 - 48.7}{1.0 \times 10^3 \times 9.8} = 5.0 \text{ mm.}$$

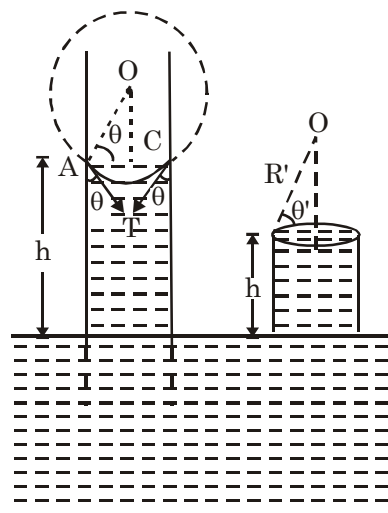
Capillary rise in a tube of insufficient length

We know, the height through which a liquid rises in the capillary tube of radius r is given by

$$\therefore h = \frac{2T}{R\rho g} \text{ or } hR = \frac{2T}{\rho g} = \text{constant}$$

When the capillary tube is cut and its length is less than h (i.e. h'). then the liquid rises up to the top of the tube and spreads in such a way that the radius (R') of the liquid meniscus increases and it becomes more flat so that $hR = h'R' = \text{Constant}$. Hence the liquid does not overflow.

$$\begin{aligned} \text{If } h' < h \text{ then } R' > R & \quad \text{or} \quad \frac{r}{\cos \theta'} > \frac{r}{\cos \theta} \\ \Rightarrow \cos \theta' < \cos \theta & \quad \Rightarrow \theta' > \theta \end{aligned}$$



Ex.10 If a 5cm long capillary tube with 0.1 mm internal diameter open at both ends is slightly dipped in water having surface tension 75 dyne cm^{-1} , state whether (i) water will rise half way in the capillary. (ii) Water will rise up to the upper end of capillary (iii) What will overflow out of the upper end of capillary. Explain your answer.

Sol. Given that surface tension of water, $T = 75 \text{ dyne/cm}$

$$\text{Radius } r = \frac{0.1}{2} \text{ mm} = 0.05 \text{ mm} = 0.005 \text{ cm}$$

$$\text{density } \rho = 1 \text{ gm/cm}^3, \text{ angle of contact, } \theta = 0^\circ$$

Let h be the height to which water rise in the capillary tube. Then

$$h = \frac{2T \cos \theta}{r\rho g} = \frac{2 \times 75 \times \cos 0^\circ}{0.005 \times 1 \times 981} \text{ cm} = 30.58 \text{ cm}$$

But length of capillary tube, $h' = 5\text{cm}$

- (i) Because $h > \frac{h'}{2}$ therefore the first possibility does not exist.
- (ii) Because the tube is of insufficient length therefore the water will rise upto the upper end of the tube.
- (iii) The water will not overflow out of the upper end of the capillary. It will rise only upto the upper end of the capillary.

The liquid meniscus will adjust its radius of curvature R' in such a way that

$$R'h' = Rh \quad \left[\because hR = \frac{2T}{\rho g} = \text{constant} \right]$$

where R is the radius of curvature that the liquid meniscus would possess if the capillary tube were of sufficient length.

$$\therefore R' = \frac{Rh}{h'} = \frac{rh}{h'} \quad \left[\because R = \frac{r}{\cos\theta} = \frac{r}{\cos 0^\circ} = r \right] = \frac{0.005 \times 30.58}{5} = 0.0306 \text{ cm}$$

Applications of surface tension

- (i) The wetting property is made use of in detergents and waterproofing. When the detergent materials are added to liquids, the angle of contact decreases and hence the wettability increases. On the other hand, when water proofing material is added to a fabric, it increases the angle of contact, making the fabric water-repellant.
- (ii) The antiseptics have very low value of surface tension. The low value of surface tension prevents the formation of drops that may otherwise block the entrance to skin or a wound. Due to low surface tension the antiseptics spreads properly over the wound. The lubricating oils and paints also have low surface tension. So they can spread properly.
- (iii) Surface tension of all lubricating oils and paints is kept low so that they spread over a large area.
- (iv) Oil spreads over the surface of water because the surface tension of oil is less than the surface tension of cold water.
- (v) A rough sea can be calmed by pouring oil on its surface.

Effect of temperature and impurities on surface tension

The surface tension of a liquid decreases with the rise in temperature and vice versa. According to

Ferguson, $T = T_0 \left(1 - \frac{\theta}{\theta_c} \right)^n$ where T_0 is surface tension at 0°C , θ is absolute temperature of the

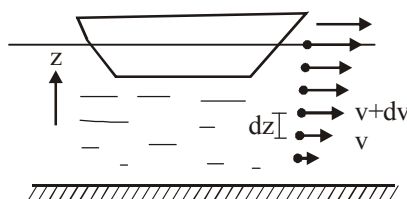
liquid, θ_c is the critical temperature and n is a constant varies slightly from liquid and has means value 1.21. This formula shows that the surface tension becomes zero at the critical temperature, where the while machinery parts get jammed in winter.

The surface tension of a liquid change appreciably with addition of impurities. For example, surface tension of water increases with addition of highly soluble substances like NaCl , ZnSO_4 etc. On the other hand surface tension of water gets reduced with addition of sparingly soluble substances like phenol, soap etc.

VISCOSITY

When a layer of a fluid slips or tends to slip on another layer in contact, the two layers exert tangential forces on each other. The directions are such that the relative motion between the layers is opposed. this property of a fluid to oppose relative motion between its layers is called viscosity. The forces between the layers opposing relative motion between them are known as the forces of viscosity. Thus, viscosity may be thought of as the internal friction of a fluid in motion.

If a solid surface is kept in contact with a fluid and is moved, forces of viscosity appear between the solid surface and the fluid layer in contact. the fluid in contact is dragged with the solid. If the viscosity is sufficient, the layer moves with the solid and there is no relative slipping. When a boat moves slowly on the water of a calm river, the water in contact with the boat is dragged with it, whereas the water in contact with the bed of the river remains at rest. Velocities of different layers are different. Let v be the velocity of the layer at a distance z from the bed and $v + dv$ be the velocity at a distance $z + dz$ (figure).



Thus, the velocity differs by dv in going through a distance dz perpendicular to it. The quantity dv/dz is called the velocity gradient.

The force of viscosity between two layers of a fluid is proportional to the velocity gradient in the direction perpendicular to the layers. Also the force is proportional to the area of the layer.

Thus, if F is the force exerted by a layer of area A on a layer in contact,

$$F \propto A \text{ and } F \propto dv/dz$$

or,

$$F = -\eta A dv/dz$$

The negative sign is included as the force is frictional in nature and opposes relative motion. The constant of proportionality η is called the coefficient of viscosity.

The SI unit of viscosity can be easily worked out from equation. It is N-s/m^2 . However, the corresponding CGS unit dyne-s/cm^2 is in common use and is called a poise in honour of the French scientist Poiseuille. We have

$$1 \text{ poise} = 0.1 \text{ N-s/m}^2$$

TERMINAL VELOCITY

The viscous force on a solid moving through a fluid is proportional to its velocity. When a solid is dropped in a fluid, the forces acting on it are

- weight W acting vertically downward,
- the viscous force F acting vertically upward and
- the buoyancy force B acting vertically upward.

The weight W and the buoyancy B are constant but the force F is proportional to the velocity v , initially, the velocity and hence the viscous force F is zero and the solid is accelerated due to the force $W - B$. Because of the acceleration, the velocity increases. Accordingly, the viscous force also increases. At a certain instant the viscous force becomes equal to $W - B$. the net force then becomes zero and the solid falls with constant velocity. This constant velocity is known as the terminal velocity.

Consider a spherical body falling through a liquid. Suppose the density of the body = ρ , density of the liquid = σ , radius of the sphere = r and the terminal velocity = v_0 . The viscous force is

$$F = 6\pi\eta r v_0$$

the weight

$$W = \frac{4}{3}\pi r^3 \rho g$$

and the buoyancy force

$$B = \frac{4}{3}\pi r^3 \sigma g$$

We have

$$6\pi\eta r v_0 = W - B = \frac{4}{3}\pi r^3 \rho g - \frac{4}{3}\pi r^3 \sigma g$$

or

$$v_0 = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

- Q.** A large wooden plate of area 10 m^2 floating on the surface of a river is made to move horizontally with a speed of 2 m/s by applying a tangential force. If the river is 1 m deep and the water in contact with the bed is stationary, find the tangential force needed to keep the plate moving. Coefficient of viscosity of water at the temperature of the river = 10^{-2} poise.

- Sol.** The velocity decreases from 2 m/s to zero in 1 m of perpendicular length. Hence, velocity gradient.

$$= dv/dx = 2 \text{ s}^{-1}$$

Now,

$$\eta = \left| \frac{F/A}{dv/dx} \right|$$

or,

$$10^{-3} \frac{\text{N} \cdot \text{s}}{\text{m}^2} = \frac{F}{(10\text{m})^2 (2\text{s}^{-1})}$$

or,

$$F = 0.02 \text{ N}.$$

- Q.** The velocity of water in a river is 18 km/hr near the surface. If the river is 5 m deep, find the shearing stress between the horizontal layers of water. The coefficient of viscosity of water = 10^{-2} poise.

- Sol.** The velocity gradient in vertical direction is

$$\frac{dv}{dx} = \frac{18 \text{ km/hr}}{5 \text{ m}} = 1.0 \text{ s}^{-1}$$

The magnitude of the force of viscosity is

$$F = \eta A \frac{dv}{dx}$$

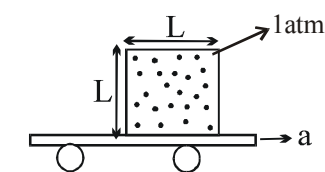
The shearing stress is

$$F/A = \eta \frac{dv}{dx} = (10^{-2} \text{ poise}) (1.0 \text{ s}^{-1}) = 10^{-3} \text{ N/m}^2$$

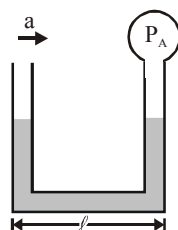
EXERCISE (S-1)

Fluid Statics

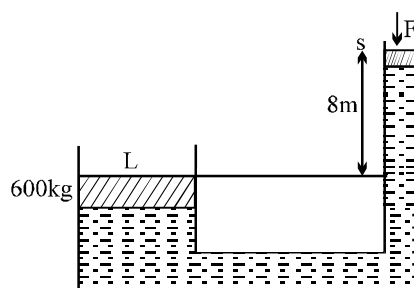
1. A spherical tank of 1.2 m radius is half filled with oil of relative density 0.8. If the tank is given a horizontal acceleration of 10 m/s^2 . Calculate the inclination of the oil surface to horizontal and maximum pressure on the tank.
2. A cubical sealed vessel with edge L is placed on a cart, which is moving horizontally with an acceleration 'a' as shown in figure. The cube is filled with an ideal fluid having density ρ . Find the gauge pressure at the centre of the cubical vessel.



3. A liquid of density ρ is filled in a U-tube, whose one end is open & at the other end a bulb is fitted whose pressure is P_A . Now this tube is moved horizontally with acceleration 'a' as shown in the figure. During motion it is found that liquid in both column is at same level at equilibrium. If atmospheric pressure is P_0 , then find the value of P_A .

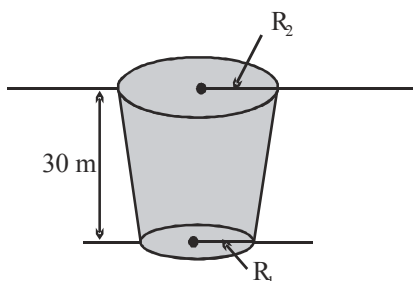


4. For the system shown in the figure, the cylinder on the left at L has a mass of 600 kg and a cross sectional area of 800 cm^2 . The piston on the right, at S, has cross sectional area 25 cm^2 and negligible weight. If the apparatus is filled with oil ($\rho = 0.75 \text{ gm/cm}^3$) Find the force F required to hold the system in equilibrium.

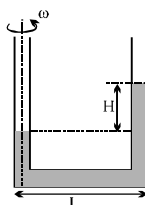


5. Dams at two different locations are needed to form a lake. When the lake is filled, the water level will be at top of both dams. The second dam is twice as high and twice as wide as the first dam. The force of the water on the second dam is how much greater than the force on the first? (Ignore atmospheric pressure since it is pushing on both sides of the dams.)

6. As the drawing illustrates, a pond full of water has the shape of an inverted cone with the tip sliced off and has a depth of 30m. The atmospheric pressure above the pond is $1.0 \times 10^5 \text{ Pa}$. The circular top surface (radius = R_2) and circular bottom surface (radius = R_1) of the pond are both parallel to the ground. The magnitude of the force acting on the top surface by the liquid is the same as the magnitude of the force acting on the bottom surface by the liquid. Obtain $\frac{R_2}{R_1}$.



7. A U tube is rotated about one of its limbs with an angular velocity ω . Find the difference in height H of the liquid (density ρ) level, where diameter of the tube $d \ll L$. **[IIT-JEE 2005]**



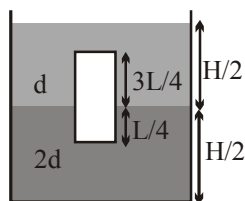
Buoyancy

8. A container of a large uniform cross-sectional area A resting on a horizontal surface holds two immiscible, non viscous and incompressible liquids of densities d and $2d$ each of height $\frac{H}{2}$ as shown.

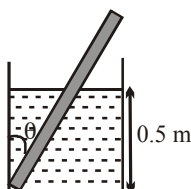
The lower density liquid is open to atmosphere. A homogeneous solid cylinder of length L ($< \frac{H}{2}$),

cross-sectional area $\frac{A}{5}$ is immersed such that it floats with its axis vertical to the liquid-liquid interface

with length $\frac{L}{4}$ in denser liquid. Find the density of the solid cylinder.



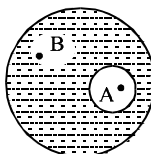
9. In air an object weighs 15N, when immersed completely in water the same object weighs 12N. When immersed in another liquid completely, it weighs 13N. Find
(a) the specific gravity of the object and
(b) the specific gravity of the other liquid.
10. A solid sphere of radius R is floating in a liquid of density ρ with half of its volume submerged. If the sphere is slightly pushed and released, it starts performing simple harmonic motion. Find the frequency of these oscillations. [IIT-JEE 2004]
11. A wooden plank of length 1 m and uniform cross-section is hinged at one end to the bottom of a tank as shown. The tank is filled with water upto a height of 0.5 m. The specific gravity of the plank is 0.5. Find the angle θ made by the plank in equilibrium position



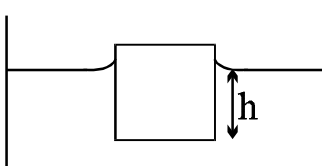
12. The volume of an air bubble is doubled as it rises from the bottom of a lake to its surface. If the atmospheric pressure is H m of mercury & the density of mercury is n times that of lake water. Find the depth of the lake.

Surface Tension

13. There is an air bubble of radius R inside a drop of water of radius $3R$. Find the ratio of gauge pressure at point A to the gauge pressure at point B.



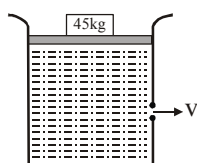
14. Two arms of a U-tube have unequal diameters $d_1 = 1.0$ mm and $d_2 = 1.0$ cm. If water (surface tension 7×10^{-2} N/m) is poured into the tube held in the vertical position, find the difference of level of water in the U-tube. Assume the angle of contact to be zero.
15. A cube with a mass 'm' completely wettable by water floats on the surface of water. Each side of the cube is 'a'. What is the distance h between the lower face of cube and the surface of the water if surface tension is S . Take density of water as ρ_w . Take angle of contact as zero.



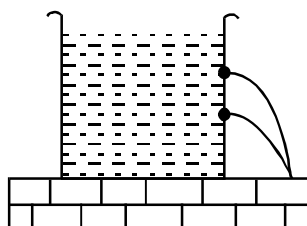
Fluid Dynamics

16. A jet of water having velocity = 10 m/s and stream cross-section = 2 cm^2 hits a flat plate perpendicularly, with the water splashing out parallel to plate. Find the force that the plate experiences.

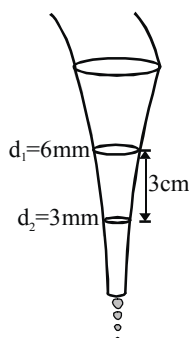
17. A laminar stream is flowing vertically down from a tap of cross-section area 1 cm^2 . At a distance 10 cm below the tap, the cross-section area of the stream has reduced to $1/2 \text{ cm}^2$. Find the volumetric flow rate of water from the tap.
18. Calculate the rate of flow of glycerine of density $1.25 \times 10^3 \text{ kg/m}^3$ through the conical section of a pipe if the radii of its ends are 0.1 m & 0.04 m and the pressure drop across its length is 10 N/m^2 .
19. A cylindrical vessel open at the top is 20 cm high and 10 cm in diameter. A circular hole whose cross-sectional area 1 cm^2 is cut at the centre of the bottom of the vessel. Water flows from a tube above it into the vessel at the rate $100 \text{ cm}^3 \text{ s}^{-1}$. Find the height of water in the vessel under steady state.
20. A large cylindrical tank of cross-sectional area 1 m^2 is filled with water. It has a small hole at a height of 1 m from the bottom. A movable piston of mass 5 kg is fitted on the top of the tank such that it can slide in the tank freely. A load of 45 kg is applied on the top of water by piston, as shown in figure. Find the value of v when piston is 7 m above the bottom ($g = 10 \text{ m/s}^2$)



21. In a cylindrical vessel containing liquid of density ρ , there are two holes in the side walls at heights h_1 and h_2 respectively such that the range of efflux at the bottom of the vessel is same. Find the height of a hole, for which the range of efflux would be maximum.

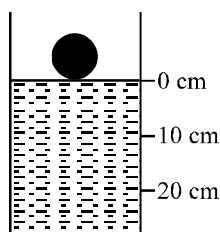


22. The tap in the garden was closed in appropriately resulting in the water flowing freely out of it which forms a downward narrowing beam. The beam of water has a circular cross-section, the diameter of the circle is 6 mm at one point and 3 cm below it is only 3 mm as shown in figure. If the rate of water wasted is $(x \times \pi) \text{ mL/minute}$ then find the value of x . (Neglect the effect of viscosity and surface tension of the flowing water.)

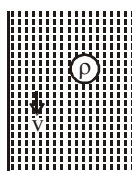


23. A spherical ball of radius 1×10^{-4} m and density 10^4 kg/m³ falls freely under gravity through a distance h before entering a tank of water. If after entering the water the velocity of the ball does not change, find h . The viscosity of water is 9.8×10^{-6} N-s/m².

24. A spherical ball of density ρ and radius 0.003m is dropped into a tube containing a viscous fluid filled up to the 0 cm mark as shown in the figure. Viscosity of the fluid = 1.260 N.m⁻² and its density $\rho_L = \rho/2 = 1260$ kg.m⁻³. Assume the ball reaches a terminal speed by the 10 cm mark. Find the time taken by the ball to traverse the distance between the 10 cm and 20 cm mark. (g = acceleration due to gravity = 10 ms⁻²)



- 25.** A liquid of density $\rho = 1000 \text{ kg/m}^3$ and coefficient of viscosity $\eta = 0.1 \text{ Ns/m}^2$ is flowing down in vertical pipe of large cross section. A small ball of density $\rho_0 = 100 \text{ kg/m}^3$ and $r = 5 \text{ cm}$ will be at rest in flowing liquid, if velocity of flowing liquid is 10 k m/s . Then find the value of k .

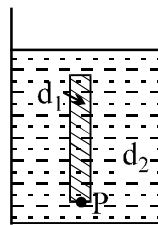


- 26.** A small sphere falls from rest in a viscous liquid. Due to friction, heat is produced. Find the relation between the rate of production of heat and the radius of the sphere at terminal velocity.

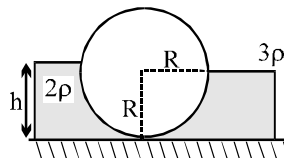
[IIT-JEE 2004]

EXERCISE (S-2)

1. Place a glass beaker, partially filled with water, in a sink. The beaker has a mass 390 gm and an interior volume of 500cm^3 . You now start to fill the sink with water and you find, by experiment, that if the beaker is less than half full, it will float; but if it is more than half full, it remains on the bottom of the sink as the water rises to its rim. What is the density of the material of which the beaker is made?
2. A thin rod of length L & area of cross-section S is pivoted at its lowest point P inside a stationary, homogeneous & non-viscous liquid (Figure). The rod is free to rotate in a vertical plane about a horizontal axis passing through P . The density d_1 of the material of the rod is smaller than the density d_2 of the liquid. The rod is displaced by a small angle θ from its equilibrium position and then released. Show that the motion of the rod is simple harmonic and determine its angular frequency in terms of the given parameters.

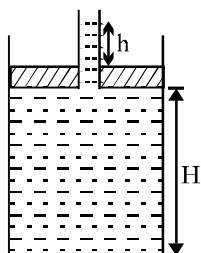


3. A hollow cone floats with its axis vertical upto one-third of its height in a liquid of relative density 0.8 and with its vertex submerged. When another liquid of relative density ρ is filled in it upto one-third of its height, the cone floats upto half its vertical height. The height of the cone is 0.10 m and the radius of the circular base is 0.05 m. Find the specific gravity ρ .
4. In the figure shown, the heavy cylinder (radius R) resting on a smooth surface separates two liquids of densities 2ρ and 3ρ . Find the height 'h' for the equilibrium of cylinder.

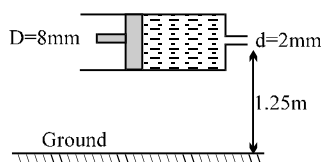


5. A small wooden ball of density ' ρ ' is immersed in water of density σ to depth h and then released. The height H above the surface of water up to which the ball jumps out of water is $\left(\frac{\sigma}{\rho} - \frac{n}{2}\right)h$. Value of n is (Neglect the effect of viscosity and surface tension and assume all quantities in SI unit)

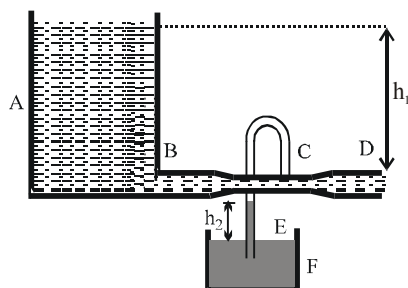
6. A piston of mass $M = 3\text{ kg}$ and radius $R = 4\text{ cm}$ has a hole into which a thin pipe of radius $r = 1\text{ cm}$ is inserted. The piston can enter a cylinder tightly and without friction, and initially it is at the bottom of the cylinder. 750 gm of water is now poured into the pipe so that the piston & pipe are lifted up as shown. Find the height H of water in the cylinder and height h of water in the pipe.



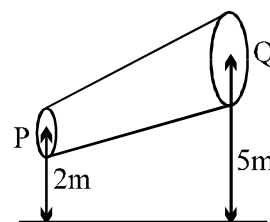
7. A cylindrical wooden float whose base area is 4 m^2 and height 1 m drifts on the water surface in vertical position. Density of wood is 500 kg/m^3 and that of water is 1000 kg/m^3 . What minimum work (in kJ) must be performed to take the float out of the water?
8. Consider a horizontally oriented syringe containing water located at a height of 1.25 m above the ground. The diameter of the plunger is 8 mm and the diameter of the nozzle is 2 mm . The plunger is pushed with a constant speed of 0.25 m/s . Find the horizontal range of water stream on the ground. Take $g = 10\text{ m/s}^2$. [IIT-JEE 2004]



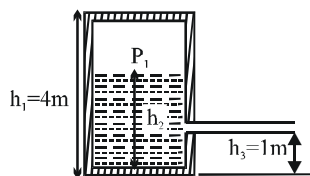
9. Two very large open tanks A and F both contain the same liquid. A horizontal pipe BCD, having a constriction at C leads out of the bottom of tank A, and a vertical pipe E opens into the constriction at C and dips into the liquid in tank F. Assume streamline flow and no viscosity. If the cross section at C is one half that at D and if D is at a distance h_1 below the level of liquid in A, to what height h_2 (in terms of h_1) will liquid rise in pipe E?



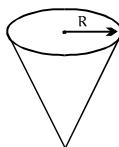
10. A non-viscous liquid of constant density 1000 kg/m^3 flows in a streamline motion along a tube of variable cross section. The tube is kept inclined in the vertical plane as shown in the figure. The area of cross section of the tube at two points P and Q at heights of 2 meters and 5 meters are respectively $4 \times 10^{-3}\text{ m}^2$ and $8 \times 10^{-3}\text{ m}^2$. The velocity of the liquid at point P is 1 m/s . Find the work done per unit volume by the pressure and the gravity forces as the fluid flows from point P to Q.



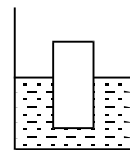
11. A large tank of (height $h_1 = 4\text{m}$) water has a hose connected to it, as shown in figure. The tank is sealed at the top and has compressed air between the water surface & the top. When the water height h_2 is 3m, the gauge pressure of air $P_1 = 1 \times 10^5 \text{N/m}^2$. Assume that the air above the water surface expands isothermally, the velocity of flow out of the hose when h_2 has decreased to 2m is given by $\alpha \text{ m/s}$? Fill $\frac{\alpha^2}{5}$ in OMR sheet. (Assume ideal fluid flow & $P_{\text{atm}} = 10^5 \text{N/m}^2$)



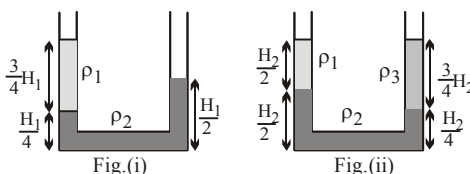
12. Two large vertical parallel plates separated by a gap of $d = 1\text{mm}$ have a highly viscous liquid of density 800 kg/m^3 and viscosity coefficient $\eta = 5 \text{ poise}$ flowing steadily under gravity in between the gap. Find the velocity gradient of flow near plates surface in sec^{-1} . (1 poise = 0.1 NS/m^2).
13. An inverted hollow cone, of height H and radius at the top $= R$, is filled completely with a liquid of density d . Neglect atmosphere pressure.
- Find horizontal component of the force which the liquid exerts on half curved side wall of the cone.
 - Find the vertical downward component of the thrust force due to water pressure which the liquid exerts on half side curved wall of the cone.



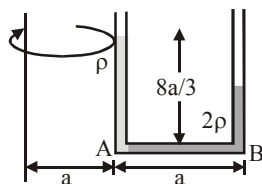
14. A liquid of density 2ρ is filled in a cylindrical vessel, whose cross-sectional area is $2A$. A wooden cylinder of height H , cross-sectional area A and density ρ is floating in the liquid at equilibrium with its axis vertical. The cylinder is pushed down by a small distance x from its equilibrium position and released. Find its initial acceleration.



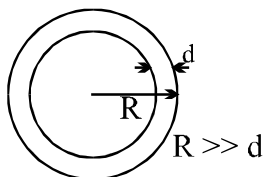
15. A U-tube filled with a liquid of volumetric coefficient of $10^{-5}/^\circ\text{C}$ lies in a vertical plane. The height of liquid column in the left vertical limb is 100 cm. The liquid in the left vertical limb is maintained at a temperature $= 0^\circ\text{C}$ while the liquid in the right limb is maintained at a temperature $= 100^\circ\text{C}$. Find the difference in levels in the two limbs.
16. There are three immiscible liquids of densities ρ_1 , ρ_2 and ρ_3 . If the first two liquids are taken in a U-tube then they stand like fig(i). When all the three liquids are taken in U-tube, they stand like fig(ii). Find the ratio of the densities of the three liquids.



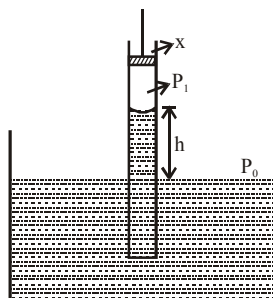
17. The interface of two liquids of densities ρ and 2ρ respectively lies at the point A in a U tube at rest. The height of liquid column above A is $8a/3$ where $AB = a$. The cross sectional area of the tube is S. With what angular velocity the tube must be whirled about a vertical axis at a distance 'a' such that the interface of the liquids shifts towards B by $2a/3$.



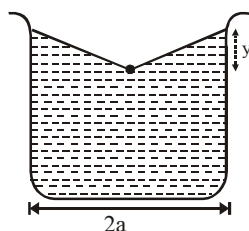
18. An expansible balloon filled with air floats on the surface of a lake with $2/3$ of its volume submerged. How deep must it be sunk in the water so that it is just in equilibrium neither sinking further nor rising? It is assumed that the temperature of the water is constant & that the height of the water barometer is 9 meters.
19. A soap bubble has radius R and thickness d ($d \ll R$) as shown. It collapses into a spherical drop. Find the ratio of excess pressure in the drop to the excess pressure inside the bubble.



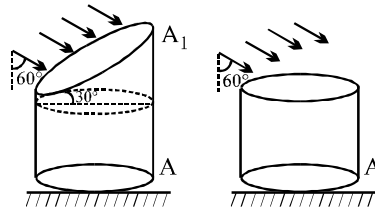
20. A capillary of radius $r = 0.2$ mm is dipped vertically in a liquid of density, ($\rho = 1 \text{ gm/cc}$). A small piston (x) is inserted inside the capillary which maintains the constant pressure of air ($P_1 = 200 \text{ dynes/cm}^2$) above the hemispherical meniscus. If ambient pressure is $P_0 = 1000 \text{ dynes/cm}^2$ and surface tension of liquid is $T = 72 \text{ dynes/cm}$, then find the height h of liquid in cm. (weight of curved part of liquid is ignored)



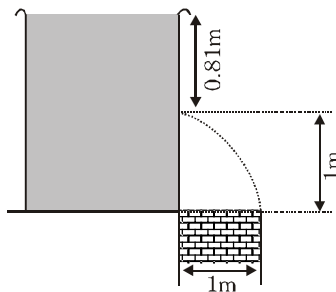
21. A container of width $2a$ is filled with a liquid. A thin wire of weight per unit length λ is gently placed over the liquid surface in the middle of the surface as shown in the figure. As a result, the liquid surface is depressed by a distance y ($y \ll a$). Determine the surface tension of the liquid. [2004, 2M]



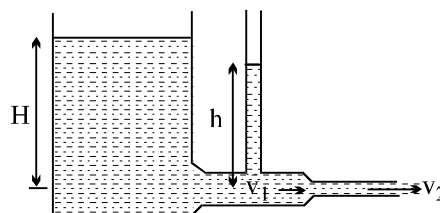
22. A vertical cylindrical container of base area A and upper cross-section area A_1 making an angle 30° with the horizontal is placed in an open rainy field as shown near another cylindrical container having same base area A . Find the ratio of rates of collection of water in the two containers.



23. A wide vessel with small hole in the bottom is filled with water and kerosene. Neglecting viscosity, find the velocity of water flow v , if the thickness of water layer is h_1 and that of kerosene layer is h_2 is (density of water ρ_1 gm/cc and that of kerosene is ρ_2 gm/cc).
24. For the arrangement shown in the figure. The time interval in seconds after which the water jet ceases to cross the wall is found to be $(\alpha)^3$. Area of the cross section of the tank $A = \sqrt{5} \text{ m}^2$ and area of the orifice $a = 32 \text{ cm}^2$. [Assume that the container remaining fixed]. Find the value of α . (Take $g = 10 \text{ m/s}^2$)

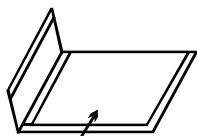


25. Water fills a reservoir, open to the atmosphere, to height $H = 5\text{m}$ above the centerline of a horizontal exit pipe at the bottom of the reservoir. The first section of the pipe has radius $r_1 = 2 \text{ cm}$, the unknown velocity there is v_1 and this section of the pipe has a manometer in which the water rises to an unknown height h above the centerline of the pipe. The water leaves the second section of radius $r_2 = 1 \text{ cm}$, with speed v_2 . Assume the flow is incompressible, frictionless, irrotational, and steady. Also, assume the reservoir is so large compared to the pipe, that the water level in the reservoir is almost constant.
- What is the speed, v_2 , of the water leaving the exit pipe?
 - What is the speed, v_1 , in the first section of the pipe?
 - What is the height, h , of the water in the manometer?
 - What is the volume flow rate of the water flowing through the exit pipe?

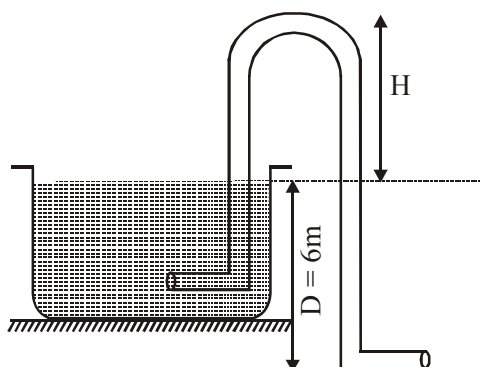


26. A glass plate of area A and mass m is hinged along one of its sides. At what speed the air should be blown parallel to its upper surface to hold the plate horizontal? The density of air is ρ . Give the

answer in term of $\sqrt{\frac{mg}{18\rho A}}$



27. Figure shows a siphon, the vessel area is very large as compared to cross section of tube. Tube has a uniform cross section, its lower end is 6m below the surface of water. What is maximum height H (in m) of the upper end for siphon to work? Take density of water = 10^3 kg/m^3 , atmospheric pressure = 10^5 N/m^2 .

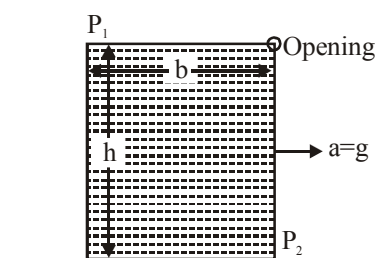


EXERCISE (O-1)

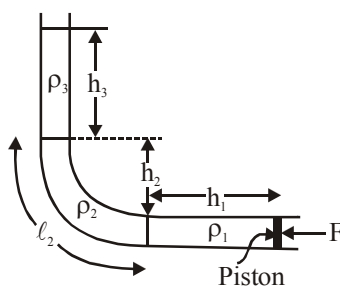
SINGLE CORRECT TYPE QUESTIONS

Fluid Statics

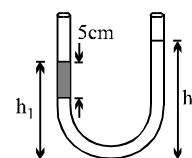
1. A vessel filled with a liquid of density ρ falls vertically downwards with an acceleration $a(<g)$. The gauge pressure P at depth h below the free surface of liquid is :
 (A) $P = h\rho(g+a)$ (B) $P = h\rho g$ (C) $P = h\rho(g-a)$ (D) $P = h\rho a$
2. A closed rectangular vessel completely filled with a liquid of density ρ moves with an acceleration $a = g$. The value of the pressure difference ($P_1 - P_2$) is :



- (A) ρgb (B) $\frac{\rho g(b+h)}{2}$ (C) $\rho(ab - gh)$ (D) ρgh
3. A tube is bent into L shape and kept in a vertical plane. If these three liquids are kept in equilibrium by the piston of area A , the value of $\frac{F}{A}$ is :



- (A) $(\rho_1 h_1 + \rho_2 h_2 + \rho_3 h_3)g$ (B) $(\rho_1 h_1 + \rho_2 l_2 + \rho_3 h_3)g$
 (C) $(\rho_2 h_2 + \rho_3 h_3)g$ (D) $(\rho_2 l_2 + \rho_3 h_3)g$
4. An open-ended U-tube of uniform cross-sectional area contains water (density $1.0 \text{ gram/centimeter}^3$) standing initially 20 centimeters from the bottom in each arm. An immiscible liquid of density $4.0 \text{ grams/centimeter}^3$ is added to one arm until a layer 5 centimeters high forms, as shown in the figure above. What is the ratio h_2/h_1 of the heights of the liquid in the two arms?

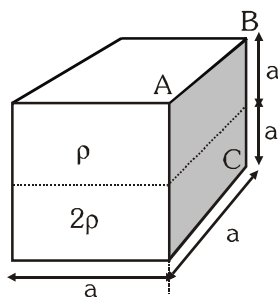


- (A) $3/1$ (B) $5/2$ (C) $2/1$ (D) $3/2$

5. A tube of length L is filled completely with incompressible liquid of mass M and closed at both the ends. The tube is then rotated in a horizontal plane about one of its ends with a uniform angular velocity ω . The force exerted by the liquid on the tube at other end is :-

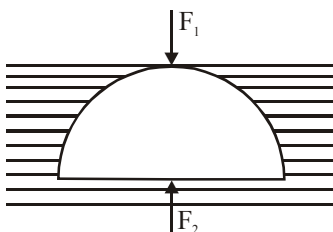
(A) $\frac{M\omega^2 L}{2}$ (B) $M\omega^2 L$ (C) $\frac{M\omega^2 L}{4}$ (D) $\frac{3M\omega^2 L}{4}$

6. A cuboid ($a \times a \times 2a$) is filled with two immiscible liquids of density 2ρ & ρ as shown in the figure. Neglecting atmospheric pressure, ratio of force on base & side wall of the cuboid is :-



(A) 2 : 3 (B) 1 : 3 (C) 5 : 6 (D) 6 : 5

7. A solid hemisphere is just pressed below the liquid, the value of $\frac{F_1}{F_2}$ is (where F_1 and F_2 are the hydrostatic forces acting on the curved and flat surfaces of the hemisphere) (Neglect atmospheric pressure).



(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) none of these

8. A thin walled cylindrical metal vessel of linear coefficient of expansion $10^{-3} \text{ } ^\circ\text{C}^{-1}$ contains benzene of volume expansion coefficient $10^{-3} \text{ } ^\circ\text{C}^{-1}$. If the vessel and its contents are now heated by 10°C , the pressure due to the liquid at the bottom.

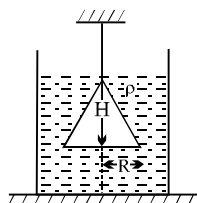
(A) increases by 2% (B) decreases by 1%
(C) decreases by 2% (D) remains unchanged

9. A cork of density 0.5 gcm^{-3} floats on a calm swimming pool. The fraction of the cork's volume which is under water is :-

(A) 0% (B) 25% (C) 10% (D) 50%

Buoyancy

10. A cone of radius R and height H , is hanging inside a liquid of density ρ by means of a string as shown in the figure. The force, due to the liquid acting on the slant surface of the cone is (Neglect atmosphere pressure)



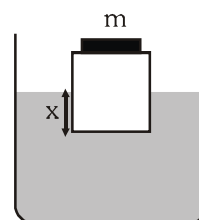
- (A) $\rho\pi gHR^2$ (B) $\pi\rho HR^2$ (C) $\frac{4}{3}\pi\rho gHR^2$ (D) $\frac{2}{3}\pi\rho gHR^2$
11. A cubical block ($a \times a \times a$), with a coin of mass ' m ' over it is floating over a liquid of density ρ . In this case x_1 depth of the block is immersed. Now the coin is removed & it is found that x_2 depth is immersed in liquid. Value of $(x_1 - x_2)$ is :-

(A) $\frac{m}{\rho a^2}$

(B) $\frac{\rho a^4}{m}$

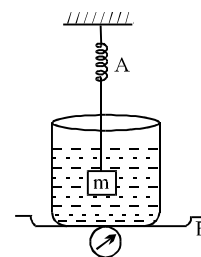
(C) $\frac{m}{2\rho a^2}$

(D) data insufficient

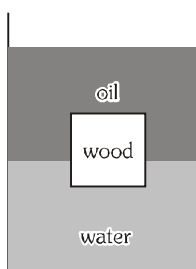


12. The spring balance A reads 2 kg with a block m suspended from it. A balance B reads 5 kg when a beaker with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid in the beaker as shown in the figure in this situation:

- (A) the balance A will read more than 2 kg
 (B) the balance B will read more than 5 kg
 (C) the balance A will read less than 2 kg and B will read less than 5 kg
 (D) the balances A and B will read 2 kg and 5 kg respectively.



13. A cubical block of wood of side of 10 cm, floats at the interface between oil and water as shown in figure with its lower face 2 cm below the interface. The density of oil is 0.6 g/cm^3 . The mass of the block is



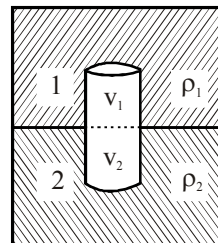
(A) 600 g

(B) 680 g

(C) 800 g

(D) 200 g

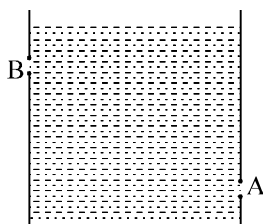
14. A solid floats with $\frac{2}{3}$ of its volume immersed in a liquid and with $\frac{3}{4}$ of its volume immersed in another liquid. What fraction of its volume will be immersed if it floats in a homogenous mixture formed of equal volumes of the liquids?
 (A) $\frac{6}{7}$ (B) $\frac{8}{11}$ (C) $\frac{11}{16}$ (D) $\frac{12}{17}$
15. A cuboidal piece of wood has dimensions a , b and c . Its relative density is d . It is floating in a large body of water such that side a is vertical. It is pushed down a bit and released. The time period of SHM executed by it is :
 (A) $2\pi\sqrt{\frac{abc}{g}}$ (B) $2\pi\sqrt{\frac{g}{da}}$ (C) $2\pi\sqrt{\frac{bc}{dg}}$ (D) $2\pi\sqrt{\frac{da}{g}}$
16. A small ball of relative density 0.8 falls into water from a height of 2m. The depth to which the ball will sink is (neglect viscous forces):
 (A) 8 m (B) 2 m (C) 6 m (D) 4 m
17. An object of density d_0 kept deep inside water of density d_w and released. During the time it moves a vertical distance h with in the water :-
 (A) The gravitational potential energy of the water in the vessel increases if $d_0 < d_w$
 (B) The gravitational potential energy of the water in the vessel decreases if $d_0 < d_w$
 (C) The gravitational potential energy of the object increases if $d_0 > d_w$
 (D) The gravitational potential energy of the object decreases if $d_0 < d_w$
18. **Statement-1 :** When a body floats such that it's parts are immersed into two immiscible liquids then force exerted by liquid-1 is of magnitude $\rho_1 v_1 g$.
Statement-2 : Total Buoyant force = $\rho_1 v_1 g + \rho_2 v_2 g$
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
19. **Statement-1 :** Submarine sailors are advised that they should not allow it to rest on floor of the ocean.
Statement-2 : The force exerted by a liquid on a submerged body may be downwards.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.



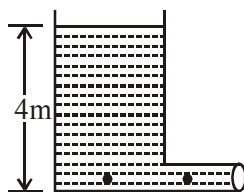
Bernoulli

20. Water is flowing steadily through a horizontal tube of non uniform cross-section. If the pressure of water is $4 \times 10^4 \text{ N/m}^2$ at a point where cross-section is 0.02 m^2 and velocity of flow is 2 m/s , what is pressure at a point where cross-section reduces to 0.01 m^2 .
 (A) $1.4 \times 10^4 \text{ N/m}^2$ (B) $3.4 \times 10^4 \text{ N/m}^2$ (C) $2.4 \times 10^{-4} \text{ N/m}^2$ (D) none of these

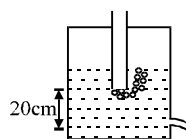
21. In the case of a fluid, Bernoulli's theorem expresses the application of the principle of conservation of:-
 (A) linear momentum (B) energy
 (C) mass (D) angular momentum
22. Two water pipes P and Q having diameters $2 \times 10^{-2} \text{ m}$ and $4 \times 10^{-2} \text{ m}$, respectively, are joined in series with the main supply line of water. The velocity of water flowing in pipe P is :-
 (A) 4 times that of Q (B) 2 times that of Q
 (C) $1/2$ times of that of Q (D) $1/4$ times that of Q
23. A cylinder of height 20 m is completely filled with water. The velocity of efflux of water (in ms^{-1}) through a small hole on the side wall of the cylinder near its bottom, is- [AIEEE - 2002]
 (A) 10 (B) 20 (C) 25.5 (D) 5
24. Two identical holes each of cross-sectional area 10^{-3} m^2 are made on the opposite sides of a tank containing water as shown in the figure. As the water comes out of the holes, the tank will experience a net horizontal force of 20 N. The difference in height between the holes A and B is.



- (A) 1 m (B) 0.5 m (C) 2 m (D) 0.25 m
25. A vent tank of large cross-sectional area has a horizontal pipe 0.12 m in diameter at the bottom. This holds a liquid whose density is 1500 kg/m^3 to a height of 4.0 m. Assume the liquid is an ideal fluid in laminar flow. In figure, the velocity with which fluid flows out is :-

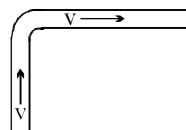


- (A) $2\sqrt{5} \text{ m/s}$ (B) $\sqrt{5} \text{ m/s}$ (C) $4\sqrt{5} \text{ m/s}$ (D) $\sqrt{10} \text{ m/s}$
26. A tube is attached as shown in closed vessel containing water. The velocity of water coming out from a small hole is :



- (A) $\sqrt{2} \text{ m/s}$ (B) 2 m/s
 (C) depends on pressure of air inside vessel (D) None of these

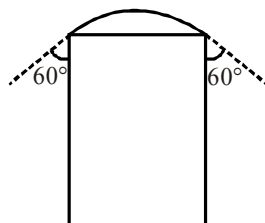
27. A fire hydrant delivers water of density ρ at a volume rate L . The water travels vertically upward through the hydrant and then does 90° turn to emerge horizontally at speed V . The pipe and nozzle have uniform cross-section throughout. The force exerted by the water on the corner of the hydrant is :-



- (A) ρVL (B) zero (C) $2\rho VL$ (D) $\sqrt{2}\rho VL$
28. A large tank is filled with water to a height H . A small hole is made at the base of the tank. It takes T_1 time to decrease the height of water to H/η , ($\eta > 1$) and it takes T_2 time to take out the rest of water. If $T_1 = T_2$, then the value of η is :
- (A) 2 (B) 3 (C) 4 (D) $2\sqrt{2}$
29. Water is filled in a container upto height 3m. A small hole of area 'a' is punched in the wall of the container at a height 52.5 cm from the bottom. The cross sectional area of the container is A. If $\frac{a}{A} = 0.1$ then v^2 is (where v is the velocity of water coming out of the hole) [IIT-JEE' 2005 (Scr)]
- (A) 48 (B) 51 (C) 50 (D) 51.5

Surface Tension

30. If two soap bubbles of different radii are connected by a tube,
- (A) air flows from the bigger bubble to the smaller bubble till the sizes become equal
 (B) air flows from bigger bubble to the smaller bubble till the sizes are interchanged
 (C) air flows from the smaller bubble to the bigger
 (D) there is no flow of air.
31. A soap bubble is being blown on a tube of radius 1 cm. The surface tension of the soap solution is 0.05 N/m and the bubble makes an angle of 60° with the tube as shown. The excess of pressure over the atmospheric pressure in the tube is :

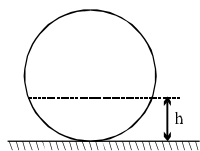


- (A) 5 Pa (B) 1 Pa (C) 10 Pa (D) 20 Pa
32. When an air bubble rises from the bottom of a deep lake to a point just below the water surface, the pressure of air inside the bubble :-
- (A) is greater than the pressure outside it (B) is less than the pressure outside it
 (C) increases as the bubble moves up (D) remains same as the bubble moves up

33. Two mercury drops (each of radius 'r') merge to form a bigger drop. The surface energy of the bigger drop, if $\frac{1}{\pi}$ is the surface tension (in SI unit), is :

(A) $2^{5/3} r^2$ (B) $4r^2$ (C) $2r^2$ (D) $2^{8/3} r^2$

34. A liquid is filled in a spherical container of radius R till a height h. At this position the liquid surface at the edges is also horizontal. The contact angle is :-



(A) 0 (B) $\cos^{-1}\left(\frac{R-h}{R}\right)$ (C) $\cos^{-1}\left(\frac{h-R}{R}\right)$ (D) $\sin^{-1}\left(\frac{R-h}{R}\right)$

35. An open capillary tube is lowered in a vessel with mercury. The difference between the levels of the mercury in the vessel and in the capillary tube $\Delta h = 4.6\text{mm}$. What is the radius of curvature of the mercury meniscus in the capillary tube? Surface tension of mercury is 0.46 N/m , density of mercury is 13.6 gm/cc .

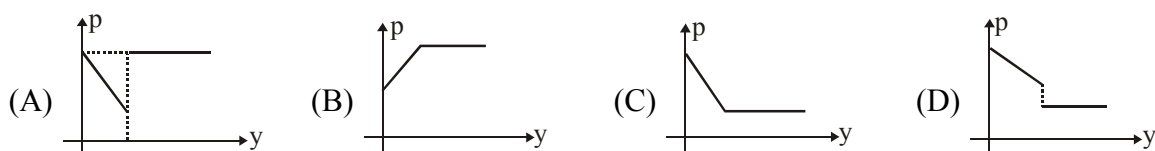
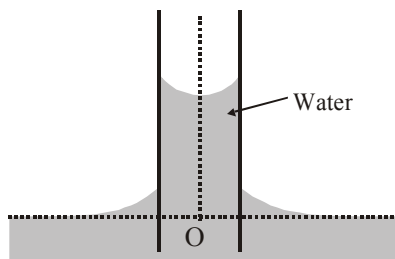
(A) $\frac{1}{340}\text{ m}$ (B) $\frac{1}{680}\text{ m}$
(C) $\frac{1}{1020}\text{ m}$ (D) Information insufficient

36. A container, whose bottom has round holes with diameter 0.1 mm is filled with water. The maximum height in cm upto which water can be filled without leakage will be what?

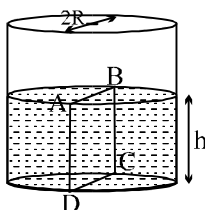
Surface tension $= 75 \times 10^{-3}\text{ N/m}$ and $g = 10\text{ m/s}^2$:

(A) 20 cm (B) 40 cm (C) 30 cm (D) 60 cm

37. Water rises in a capillary as shown. The correct graph of pressure (p) vs height (y) above point O along the axis of capillary is :-



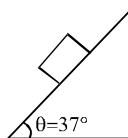
38. If two soap bubbles of different radii are connected by a tube- [AIEEE - 2004]
 (A) air flows from the bigger bubble to the smaller bubble till the sizes become equal
 (B) air flows from bigger bubble to the smaller bubble till the sizes are interchanged
 (C) air flows from the smaller bubble to the bigger bubble
 (D) there is no flow of air
39. A 20 cm long capillary tube is dipped in water. The water rises upto 8 cm. If the entire arrangement is put in a freely falling elevator, the length of water column in the capillary tube will be-
 (A) 8 cm (B) 10 cm (C) 4 cm (D) 20 cm
40. Water is filled up to a height h in a beaker of radius R as shown in the figure. The density of water is ρ , the surface tension of water is T and the atmospheric pressure is P_0 . Consider a vertical section ABCD of the water column through a diameter of the beaker. The force on water on one side of this section by water on the other side of this section has magnitude [IIT-JEE 2007]



- (A) $|2P_0Rh + \pi R^2 \rho gh - 2RT|$ (B) $|2P_0Rh + R\rho gh^2 - 2RT|$
 (C) $|P_0\pi R^2 + R\rho gh^2 - 2RT|$ (D) $|P_0\pi R^2 + R\rho gh^2 + 2RT|$

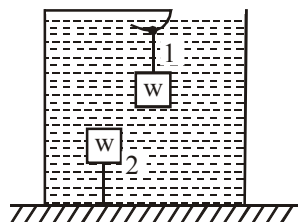
Viscosity

41. Two drops of same radius are falling through air with steady velocity of v cm/s. If the two drops coalesce, what would be the terminal velocity?
 (A) $4v$ (B) $(4)^{1/3}v$ (C) $2v$ (D) $64v$
42. Spherical balls of radius R are falling in a viscous fluid of viscosity η with a velocity v . the retarding viscous force acting on the spherical ball is- [AIEEE - 2004]
 (A) directly proportional to R but inversely proportional to v
 (B) directly proportional to both radius R and velocity v
 (C) inversely proportional to both radius R and velocity v
 (D) inversely proportional to R but directly proportional to velocity v
43. A cubical block of side 'a' and density ' ρ ' slides over a fixed inclined plane with constant velocity ' v '. There is a thin film of viscous fluid of thickness 't' between the plane and the block. Then the coefficient of viscosity of the thin film will be:



- (A) $\frac{3\rho a g t}{5v}$ (B) $\frac{4\rho a g t}{5v}$ (C) $\frac{\rho a g t}{v}$ (D) none of these

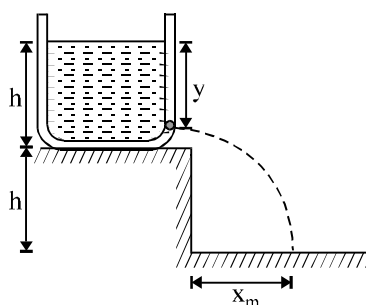
50. An iron block and a wooden block are positioned in a vessel containing water as shown in the figure. The iron block (1) hangs from a massless string with a rigid support from the top while the wooden block (2) floats being tied to the bottom through a massless string. If now the vessel starts acceleration towards right.



- (A) iron block gets deflected towards right. (B) wooden block gets deflected towards right.
(C) iron block gets deflected towards left. (D) wooden block gets deflected towards left.
51. A beaker is filled in with water is accelerated $a \text{ m/s}^2$ in $+x$ direction. The surface of water shall make an angle
(A) $\tan^{-1}(a/g)$ backwards (B) $\tan^{-1}(a/g)$ forwards
(C) $\cot^{-1}(g/a)$ backwards (D) $\cot^{-1}(g/a)$ forwards

Fluid Dynamics

52. A tank is filled upto a height h with a liquid and is placed on a platform of height h from the ground. To get maximum range x_m a small hole is punched at a distance of y from the free surface of the liquid. Then



- (A) $x_m = 2h$ (B) $x_m = 1.5 h$ (C) $y = h$ (D) $y = 0.75 h$

MATRIX MATCH TYPE QUESTION

Fluid Statics

53. Bucket A contains only water ; an identical bucket B contains water, but also contains a solid object in the water. Consider the following four situations. Which bucket weighs more ?

Column-I

- (A) The object floats in bucket B, and the buckets have the same water level
(B) The object floats in bucket B, and the buckets have the same volume of water
(C) The object sinks completely in bucket B, and the buckets have the same water level
(D) The object sinks completely in bucket B, and the buckets have the same volume of water

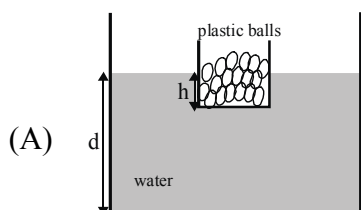
Column-II

- (P) Bucket A
(Q) Bucket B
(R) Both buckets have the same weight
(S) The answer cannot be determined from the information given.

54. Column I shows different system as describe, with some parameter while column-II gives the change in the parameter.

Column-I

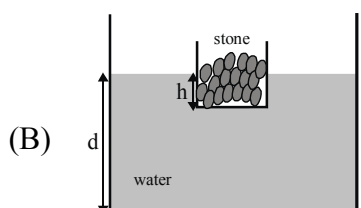
Column-II



(P) h decreases

Plastic ball is thrown from the container in the water.

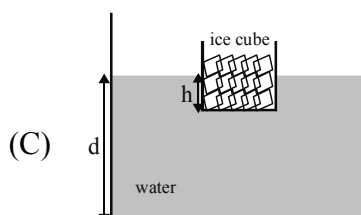
[d is the height of water level while h is the depth to which the container is submerged]



(Q) h remain same

Stone is thrown from the container in the water.

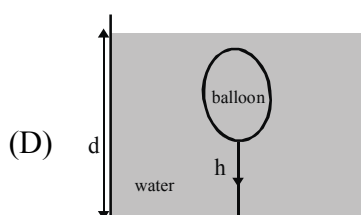
[d is the height of water level while h is the depth to which the container is submerged]



(R) h increases

Ice melts and remain in the container.

[d is the height of water level while h is the depth to which the container is submerged]



(S) d decreases

The water is heated slowly.

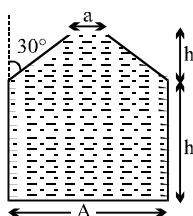
[d is the height of water level while h is the tension in the string. $\gamma_{\text{air}} > \gamma_{\text{water}}$]

(T) d remain same

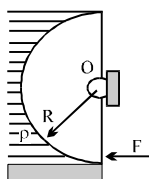
EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. The vessel shown in the figure has two sections. The lower part is a rectangular vessel with area of cross-section A and height h . The upper part is a conical vessel of height h with base area ' A ' and top area ' a ' and the walls of the vessel are inclined at an angle 30° with the vertical. A liquid of density ρ fills both the sections upto a height $2h$. Neglecting atmospheric pressure.

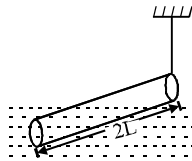


- (A) The force F exerted by the liquid on the base of the vessel is $2h\rho g \frac{(A+a)}{2}$
- (B) the pressure P at the base of the vessel is $2h\rho g \frac{A}{a}$
- (C) the weight of the liquid W is greater than the force exerted by the liquid on the base
- (D) the walls of the vessel exert a downward force $(F-W)$ on the liquid.
2. A light semi cylindrical gate of radius R is pivoted at its mid point O , of the diameter as shown in the figure holding liquid of density ρ . The force F required to prevent the rotation of the gate is equal to

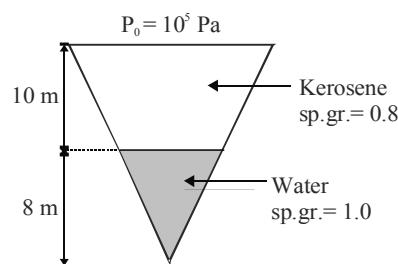


- (A) $2\pi R^3 \rho g$ (B) $2\rho g R^3 l$ (C) $\frac{2R^2 l \rho g}{3}$ (D) none of these
3. A heavy hollow cone of radius R and height h is placed on a horizontal table surface, with its flat base on the table. The whole volume inside the cone is filled with water of density ρ . The circular rim of the cone's base has a watertight seal with the table's surface and the top apex of the cone has a small hole. Neglecting atmospheric pressure find the total upward force exerted by water on the cone is
- (A) $(2/3)\pi R^2 h \rho g$ (B) $(1/3)\pi R^2 h \rho g$ (C) $\pi R^2 h \rho g$ (D) None
4. Two cubes of size 1.0 m sides, one of relative density 0.60 and another of relative density $= 1.15$ are connected by weightless wire and placed in a large tank of water. Under equilibrium the lighter cube will project above the water surface to a height of
- (A) 50 cm (B) 25 cm (C) 10 cm (D) zero

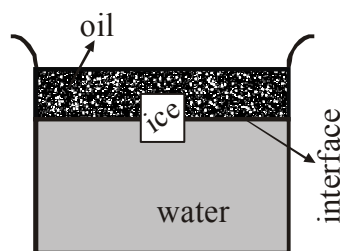
5. A slender homogeneous rod of length $2L$ floats partly immersed in water, being supported by a string fastened to one of its ends, as shown. The specific gravity of the rod is 0.75. The length of rod that extends out of water is :



- (A) L (B) $\frac{1}{2} L$ (C) $\frac{1}{4} L$ (D) $3 L$
6. Mercury is poured into a uniform vertical U-tube and water is poured in above it. The level of water is the same in both arms. A piece of wood is dropped into one arm and some water equal in weight to the piece of wood is added to the other. Then, consider the statements :
- (I) The level of mercury will be same in both the arms.
 (II) The level of water will be same in both the arms.
- (A) Both I and II are true (B) I is true but II is false
 (C) I is false but II is true (D) Both I and II are false.
7. Shown in figure, a conical container of half-apex angle 37° filled with certain quantities of kerosene and water. The force exerted by the water on the kerosene is approximately, (Take atmospheric pressure = 10^5 Pa)



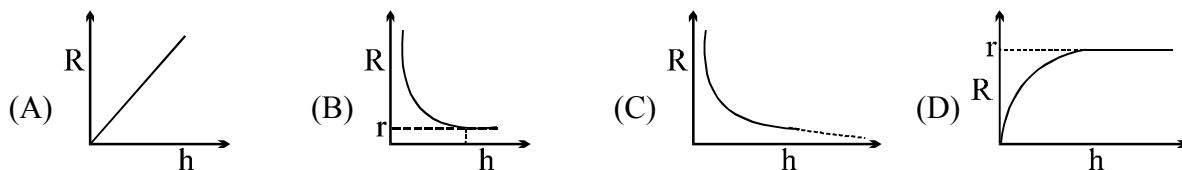
- (A) 3×10^7 N (B) 4×10^7 N (C) 2×10^7 N (D) 5×10^7 N
8. An ice cube is floating in water above which a layer of a lighter oil is poured. As the ice melts completely, the level of interface and the upper most level of oil will respectively :-



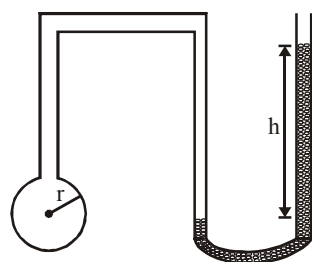
- (A) rise and fall (B) fall and rise
 (C) rise and not change (D) not change and fall

Statics

9. A long capillary tube of radius 'r' is initially just vertically completely immersed inside a liquid of angle of contact 0° . If the tube is slowly raised then relation between radius of curvature of meniscus inside the capillary tube and displacement (h) of tube can be represented by



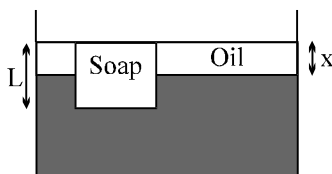
10. If the radius of the Soap-bubble on one side of tube is r and difference in height of liquid of density ρ in manometer is h, then surface tension of liquid used to make the bubble is :-



- (A) $T = 2r\rho hg$ (B) $T = \frac{r\rho g}{4}$ (C) $T = \frac{2\pi r\rho g}{2}$ (D) $T = \frac{r\rho g}{2}$

Surface tension

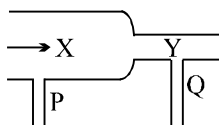
11. A rectangular bar of soap has density 800 kg/m^3 floats in water of density 1000 kg/m^3 . Oil of density 300 kg/m^3 is slowly added, forming a layer that does not mix with the water. When the top surface of the oil is at the same level as the top surface of the soap. What is the ratio of the oil layer thickness to the soap's thickness, x/L ?



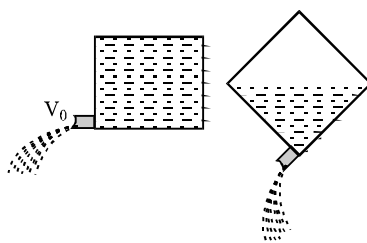
- (A) $\frac{2}{10}$ (B) $\frac{2}{7}$ (C) $\frac{3}{10}$ (D) $\frac{3}{8}$

12. Water coming out of a horizontal tube at a speed v strikes normally a vertically wall close to the mouth of the tube and falls down vertically after impact. When the speed of water is increased to $2v$.
- (A) the thrust exerted by the water on the wall will be doubled
 (B) the thrust exerted by the water on the wall will be unchanged
 (C) the energy lost per second by water striking the wall will also be four times
 (D) the energy lost per second by water striking the wall be increased eight times.

13. A steady flow of water passes along a horizontal tube from a wide section X to the narrower section Y, see figure. Manometers are placed at P and Q at the sections. Which of the statements A, B, C, D, is most correct?



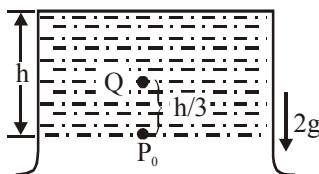
- (A) water velocity at X is greater than at Y
 (B) the manometer at P shows lower pressure than at Q
 (C) kinetic energy per m^3 of water at X = kinetic energy per m^3 at Y
 (D) the manometer at P shows greater pressure than at Q
14. A cubical box of wine has a small spout located in one of the bottom corners. When the box is full and placed on a level surface, opening the spout results in a flow of wine with a initial speed of v_0 (see figure). When the box is half empty, someone tilts it at 45° so that the spout is at the lowest point (see figure). When the spout is opened the wine will flow out with a speed of



- (A) v_0 (B) $v_0/2$ (C) $v_0/\sqrt{2}$ (D) $v_0/\sqrt[4]{2}$

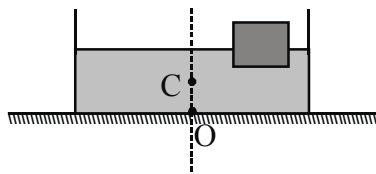
Bernoulli's

15. An inverted test tube having a liquid of density ρ accelerates down with $a = 2g$ then, $P_Q - P_0$ is:

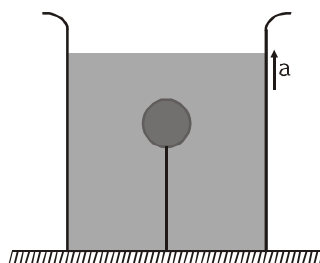


- (A) $-\frac{\rho gh}{3}$ (B) $\frac{2\rho gh}{3}$ (C) $\frac{\rho gh}{3}$ (D) indeterminate
16. Some liquid is filled in a cylindrical vessel of radius R . Let F_1 be the force applied by the liquid on the bottom of the cylinder. Now the same liquid is poured into a vessel of uniform square crss-section of side R . Let F_2 be the force applied by the liquid on the bottom of this new vessel. (Neglect atmosphere pressure) Then:
- (A) $F_1 = \pi F_2$ (B) $F_1 = \frac{F_2}{\pi}$ (C) $F_1 = \sqrt{\pi} F_2$ (D) $F_1 = F_2$

17. There is water in container with center of mass at C. Now a small wooden piece is placed towards right as shown in the figure. After putting the wooden piece.



- (A) Pressure at base remains same and centre of mass of water and wooden piece will be right of line OC.
 (B) Pressure at base remains same and centre of mass of water and wooden piece will be on line OC.
 (C) Pressure at base changes and centre of mass of water and wooden piece will be right of line OC.
 (D) Pressure at base changes and centre of mass of water and wooden piece will be on line OC.
18. The tension in a string holding a solid block below the surface of a liquid (of density greater than that of solid) as shown in figure is T_0 when the system is at rest. What will be the tension in the string if the system has an upward acceleration a ?



- (A) $T_0 \left(1 + \frac{a}{g} \right)$ (B) $T_0 \left(1 - \frac{a}{g} \right)$ (C) $2T_0 \left(1 - \frac{a}{g} \right)$ (D) $3T_0 \left(1 + \frac{2a}{g} \right)$

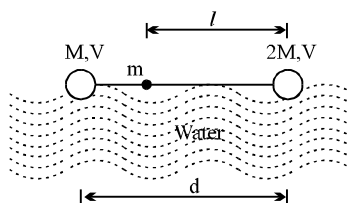
19. A wooden block floats in water in a sealed container. When the container is at rest, 25% of the block is above the water. Consider the following five situation :

- (i) the container is lifted up at constant speed.
 (ii) the container is lowered at constant speed.
 (iii) the container is lifted up at an increasing speed.
 (iv) the container is lowered at a decreasing speed.
 (v) the air pressure above the water in the container is increased.

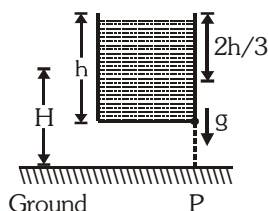
What happens in each situation?

- (A) 75% of block is submerged in situation (i) & (ii) and higher fraction is submerged in situations (iii), (iv) & (v)
 (B) 75% of block is submerged in situation (i), (ii) & (v) and higher fraction is submerged in situations (iii), (iv).
 (C) 75% of block is submerged in situation (i), (ii), (iii) & (iv) and higher fraction is submerged in situations (v).
 (D) 75% of block is submerged in all situation.

20. The frequency of a sonometer wire is f , but when the weights producing the tensions are completely immersed in water the frequency becomes $f/2$ and on immersing the weights in a certain liquid the frequency becomes $f/3$. The specific gravity of the liquid is:
- (A) $\frac{4}{3}$ (B) $\frac{16}{9}$ (C) $\frac{15}{12}$ (D) $\frac{32}{27}$
21. A dumbbell is placed in water of density ρ . It is observed that by attaching a mass m to the rod, the dumbbell floats with the rod horizontal on the surface of water and each sphere exactly half submerged as shown in the figure. The volume of the mass m is negligible. The value of length l is :-



- (A) $\frac{d(V\rho - 3M)}{2(V\rho - 2M)}$ (B) $\frac{d(V\rho - 2M)}{2(V\rho - 3M)}$ (C) $\frac{d(V\rho + 2M)}{2(V\rho - 3M)}$ (D) $\frac{d(V\rho - 2M)}{2(V\rho + 3M)}$
22. Fountains usually seen in gardens are generated by a wide pipe with an enclosure at one end having many small holes. Consider one such fountain which is produced by a pipe of internal diameter 2 cm in which water flows at a rate 3 ms^{-1} . The enclosure has 100 holes each of diameter 0.05 cm. The velocity of water coming out of the holes is (in ms^{-1}):
- (A) 0.48 (B) 96 (C) 24 (D) 48
23. An open vessel full of water is falling freely under gravity. There is a small hole in one face of the vessel, as shown in the figure. The water which comes out from the hole at the instant when hole is at height H above the ground, strikes the ground at a distance of x from P. Which of the following is correct for the situation described ?



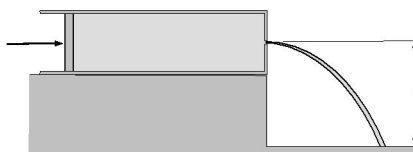
(A) The value of x is $2\sqrt{\frac{2hH}{3}}$

(B) The value of x is $\sqrt{\frac{4hH}{3}}$

(C) The value of x can't be computed from information provided.

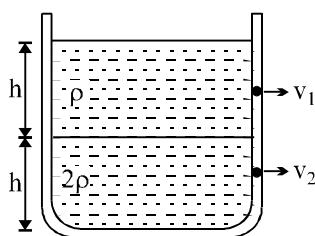
(D) The question is irrelevant as no water comes out from the hole.

24. An ideal liquid of density ρ is filled in a horizontally fixed syringe fitted with piston. There is no friction between the piston and the inner surface of the syringe. Cross-section area of the syringe is A . At one end of the syringe, an orifice of negligible cross-section area is made. When the piston is pushed into the syringe, the liquid comes out of the orifice following parabolic path and falls on the ground. With what speed the liquid strikes the ground? Neglect the air drag :-



- (A) $\sqrt{\frac{F + \rho ghA}{\rho A}}$ (B) $\sqrt{\frac{F + 2\rho ghA}{\rho A}}$ (C) $\sqrt{\frac{2F + \rho ghA}{\rho A}}$ (D) $\sqrt{\frac{2(F + \rho ghA)}{\rho A}}$

25. Equal volumes of two immiscible liquids of densities ρ and 2ρ are filled in a vessel as shown in figure. Two small holes are punched at depth $h/2$ and $3h/2$ from the surface of lighter liquid. If v_1 and v_2 are the velocities of a flux at these two holes, then v_1/v_2 is :



- (A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{\sqrt{2}}$

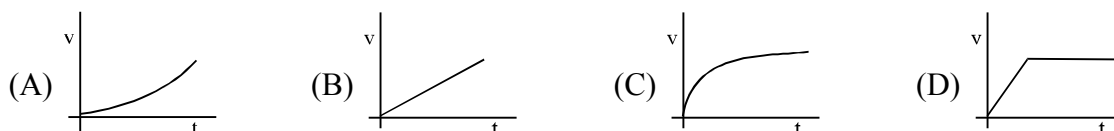
26. A vertical tank, open at the top, is filled with a liquid and rests on a smooth horizontal surface. A small hole is opened at the centre of one side of the tank. The area of cross-section of the tank is N times the area of the hole, where N is a large number. Neglect mass of the tank itself. The initial acceleration of the tank is :-

- (A) $\frac{g}{2N}$ (B) $\frac{g}{\sqrt{2}N}$ (C) $\frac{g}{N}$ (D) $\frac{g}{2\sqrt{N}}$

27. A solid metallic sphere of radius r is allowed to fall freely through air. If the frictional resistance due to air is proportional to the cross-sectional area and to the square of the velocity, then the terminal velocity of the sphere is proportional to which of the following?

- (A) r^2 (B) r (C) $r^{3/2}$ (D) $r^{1/2}$

28. Which of the following graphs best represents the motion of a raindrop?



29. Which of the following is the **incorrect** graph for a sphere falling in a viscous liquid? (Given at $t = 0$, velocity $v = 0$ and displacement $x = 0$.)



MULTIPLE CORRECT TYPE QUESTIONS

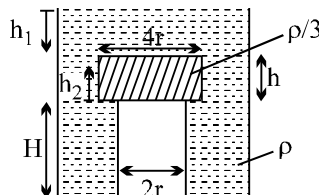
30. A cubical block is floating in a liquid with one third of its volume immersed in the liquid. When the whole system accelerates upwards with acceleration of $g/2$:-
- (A) the fraction of volume immersed in the liquid will change.
 (B) the buoyancy force on the block will change.
 (C) the buoyancy force will increase by 50 percent.
 (D) the pressure in the liquid will increased.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question no. 31 to 33

A wooden cylinder of diameter $4r$, height h and density $\rho/3$ is kept on a hole of diameter $2r$ of a tank, filled with water of density ρ as shown in the figure. The height of the base of cylinder from the base of tank is H .

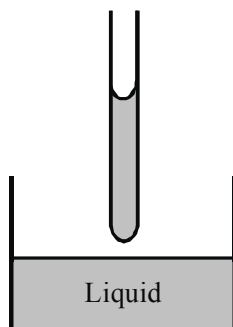
[IIT-JEE 2006]



31. If level of liquid starts decreasing slowly when the level of liquid is at a height h_1 above the cylinder, the block just starts moving up. Then, value of h_1 is :-
- (A) $\frac{2h}{3}$ (B) $\frac{5h}{4}$ (C) $\frac{5h}{3}$ (D) $\frac{5h}{2}$
32. Let the cylinder is prevented from moving up, by applying a force and water level is further decreased. Then, height of water level (h_2 in figure) for which the cylinder remains in original position without application of force is
- (A) $\frac{h}{3}$ (B) $\frac{4h}{9}$ (C) $\frac{2h}{3}$ (D) h
33. If height h_2 of water level is further decreased, then
- (A) cylinder will not move up and remains at its original position.
 (B) for $h_2 = h/3$, cylinder again starts moving up
 (C) for $h_2 = h/4$, cylinder again starts moving up
 (D) for $h_2 = h/5$ cylinder again starts moving up

Paragraph for Question no. 34 to 36

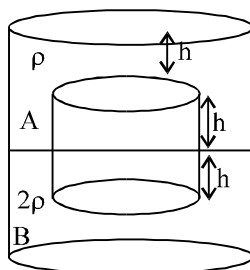
A long glass capillary (radius R) is taken out of liquid (surface tension S and density ρ) in vertical position. Angle of contact everywhere is zero. Atmospheric pressure is P_0 . It is observed that the capillary retains some liquid.



34. The radius of curvature of meniscus at the bottom of tube is
 (A) R (B) ∞ (C) $R/2$ (D) $R/3$
35. The length of liquid column retained in the capillary is
 (A) $\frac{3S}{\rho Rg}$ (B) $\frac{2S}{\rho Rg}$ (C) $\frac{4S}{\rho Rg}$ (D) zero
36. Assume equal angle of contact everywhere, the angle of contact, if the length of capillary would have been $\frac{S}{\rho Rg}$ is
 (A) $\cos^{-1}\left(\frac{1}{3}\right)$ (B) $\cos^{-1}\left(\frac{1}{2}\right)$ (C) $\cos^{-1}\left(\frac{1}{4}\right)$ (D) zero

MATRIX MATCH TYPE QUESTION

37. Shown below is a cylinder of radius R floating in vessel containing liquids A and B. Neglecting atmospheric pressure match the quantities mentioned in column-I with corresponding expression in column-II.



Column-I

- (A) Net force exerted by liquid A of density ρ on the cylinder.
 (B) Net force exerted by liquid B of density 2ρ on the cylinder.
 (C) Net force exerted by liquids A and B on the left half of the curved part of cylinder.
 (D) Net force exerted by liquid A and B on the cylinder.

Column-II

- (P) $9\rho gRh^2$
 (Q) $\pi\rho gR^2h$
 (R) $4\pi\rho gR^2h$
 (S) $3\pi\rho gR^2h$

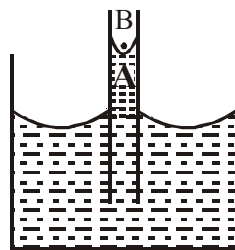
38. Capillary rise and shape of droplets on a plate due to surface tension are shown in column-II. Point A and B are just inside and outside the surface. Match the following

Column-I

Column-II

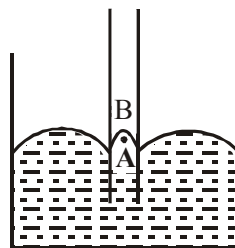
(A) liquid is wetting the solid

(P)



(B) liquid is not wetting the solid

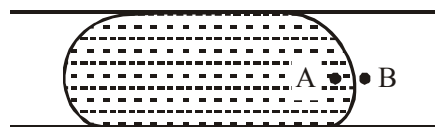
(Q)



(C) Pressure at A > pressure at B

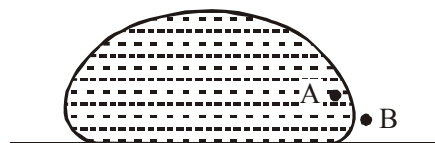
(R)

A liquid drop is pressed between two parallel glass plates.



(D) Pressure at B > Pressure at A

(S)



(T)



both end open cylindrical pipe

EXERCISE (JM)

1. A spherical solid ball of volume V is made of a material of density ρ_1 . It is falling through a liquid of density ρ_2 ($\rho_2 < \rho_1$). Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v , i.e., $F_{\text{viscous}} = -kv^2$ ($k > 0$). Then terminal speed of the ball is [AIEEE - 2008]

(1) $\sqrt{\frac{Vg(\rho_1 - \rho_2)}{k}}$ (2) $\frac{Vg\rho_1}{k}$ (3) $\sqrt{\frac{Vg\rho_1}{k}}$ (4) $\frac{Vg(\rho_1 - \rho_2)}{k}$

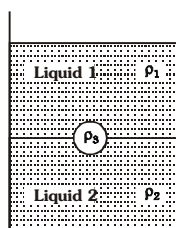
2. A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and ρ_2 , respectively. A solid ball, made of a material of density ρ_3 , is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1 , ρ_2 and ρ_3 [AIEEE - 2008]

(1) $\rho_3 < \rho_1 < \rho_2$

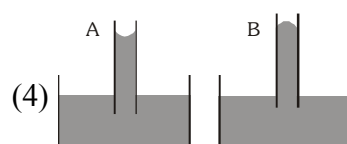
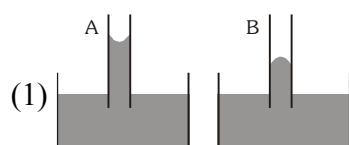
(2) $\rho_1 > \rho_3 > \rho_2$

(3) $\rho_1 < \rho_2 < \rho_3$

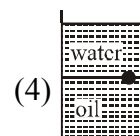
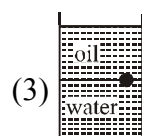
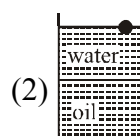
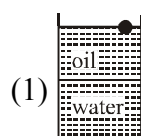
(4) $\rho_1 < \rho_3 < \rho_2$



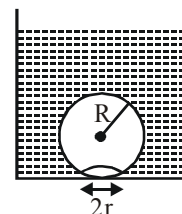
3. A capillary tube (1) is dipped in water. Another identical tube (2) is dipped in a soap-water solution. Which of the following shows the relative nature of the liquid columns in the two tubes? [AIEEE - 2008]



4. A ball is made of a material of density ρ where $\rho_{\text{oil}} < \rho < \rho_{\text{water}}$ with ρ_{oil} and ρ_{water} representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position? [AIEEE-2010]



13. On heating water, bubbles being formed at the bottom of the vessel detach and rise. Take the bubbles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel. If $r \ll R$, and the surface tension of water is T , value of r just before bubbles detach is:-



(density of water is ρ_w)

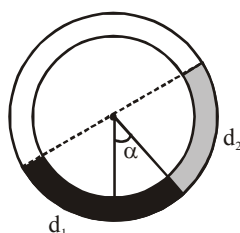
[JEE Mains-2014]

- (1) $R^2 \sqrt{\frac{\rho_w g}{T}}$ (2) $R^2 \sqrt{\frac{3\rho_w g}{T}}$ (3) $R^2 \sqrt{\frac{\rho_w g}{3T}}$ (4) $R^2 \sqrt{\frac{\rho_w g}{6T}}$

14. There is a circular tube in a vertical plane. Two liquids which do not mix and of densities d_1 and d_2 are filled in the tube. Each liquid subtends 90° angle at centre. Radius joining their interface makes

an angle α with vertical. Ratio $\frac{d_1}{d_2}$ is :

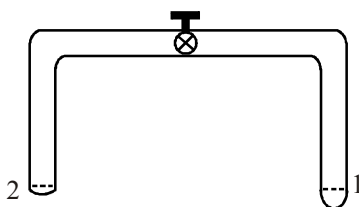
[JEE Mains-2014]



- (1) $\frac{1 + \tan \alpha}{1 - \tan \alpha}$ (2) $\frac{1 + \sin \alpha}{1 - \cos \alpha}$ (3) $\frac{1 + \sin \alpha}{1 - \sin \alpha}$ (4) $\frac{1 + \cos \alpha}{1 - \cos \alpha}$

EXERCISE (JA)

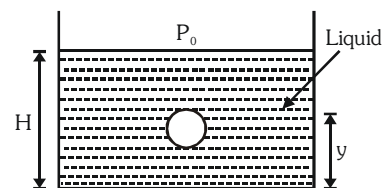
1. **STATEMENT-1** : The stream of water flowing at high speed from a garden hose pipe tends to spread like a fountain when held vertically up, but tends to narrow down when held vertically down.
and
STATEMENT-2 : In any steady flow of an incompressible fluid, the volume flow rate of the fluid remains constant. [IIT-JEE 2008]
- (A) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is a correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True
2. A glass tube of uniform internal radius (r) has a valve separating the two identical ends. Initially, the valve is in a tightly closed position. End 1 has a hemispherical soap bubble of radius r . End 2 has sub-hemispherical soap bubble as shown in figure. Just after opening the valve, [IIT-JEE 2008]



- (A) air from end 1 flows towards end 2. No change in the volume of the soap bubbles
(B) air from end 1 flows towards end 2. Volume of the soap bubble at end 1 decreases
(C) no change occurs
(D) air from end 2 flows towards end 1. Volume of the soap bubble at end 1 increases

Paragraph for Question Nos. 3 to 5

A small spherical monoatomic ideal gas bubble $\left(\gamma = \frac{5}{3}\right)$ is trapped inside a liquid of density ρ_ℓ (see figure). Assume that the bubble does not exchange any heat with the liquid. The bubble contains n moles of gas. The temperature of the gas when the bubble is at the bottom is T_0 , the height of the liquid is H and the atmospheric pressure is P_0 (Neglect surface tension). [IIT-JEE 2008]



3. As the bubble moves upwards, besides the buoyancy force the following forces are acting on it.
 (A) Only the force of gravity
 (B) The force due to gravity and the force due to the pressure of the liquid
 (C) The force due to gravity, the force due to the pressure of the liquid and the force due to viscosity of the liquid
 (D) The force due to gravity and the force due to viscosity of the liquid.
4. When the gas bubble is at a height y from the bottom, its temperature is :-
 (A) $T_0 \left(\frac{P_0 + \rho_l gH}{P_0 + \rho_l gy} \right)^{2/5}$
 (B) $T_0 \left(\frac{P_0 + \rho_l g(H-y)}{P_0 + \rho_l gH} \right)^{2/5}$
 (C) $T_0 \left(\frac{P_0 + \rho_l gH}{P_0 + \rho_l gy} \right)^{3/5}$
 (D) $T_0 \left(\frac{P_0 + \rho_l g(H-y)}{P_0 + \rho_l gH} \right)^{3/5}$
5. The buoyancy force acting on the gas bubble is (Assume R is the universal gas constant)
 (A) $p_l n R g T_0 \frac{(P_0 + \rho_l gH)^{2/5}}{(P_0 + \rho_l gy)^{7/5}}$
 (B) $\frac{\rho_l n R g T_0}{(P_0 + \rho_l gH)^{2/5} [P_0 + \rho_l g(H-y)]^{3/5}}$
 (C) $p_l n R g T_0 \frac{(P_0 + \rho_l gH)^{3/5}}{(P_0 + \rho_l gy)^{8/5}}$
 (D) $\frac{\rho_l n R g T_0}{(P_0 + \rho_l gH)^{3/5} [P_0 + \rho_l g(H-y)]^{2/5}}$
6. A cylindrical vessel of height 500 mm has an orifice (small hole) at its bottom. The orifice is initially closed and water is filled in it up to height H . Now the top is completely sealed with a cap and the orifice at the bottom is opened. Some water comes out from the orifice and the water level in the vessel becomes steady with height of water column being 200 mm. Find the fall in height (in mm) of water level due to opening of the orifice. [Take atmospheric pressure = 1.0×10^5 N/m², density of water = 1000 kg/m³ and $g = 10$ m/s². Neglect any effect of surface tension.] **[IIT-JEE-2009]**
7. Two soap bubbles A and B are kept in a closed chamber where the air is maintained at pressure 8N/m². The radii of bubbles A and B are 2 cm and 4cm, respectively. Surface tension of the soap-water used to make bubbles is 0.04 N/m. Find the ratio n_B/n_A , where n_A and n_B are the number of moles of air in bubbles A and B, respectively. [Neglect the effect of gravity] **[IIT-JEE-2009]**

Paragraph for Questions no. 8 to 10

When liquid medicine of density ρ is to be put in the eye, it is done with the help of a dropper. As the bulb on the top of the dropper is pressed, a drop forms at the opening of the dropper. We wish to estimate the size of the drop. We first assume that the drop formed at the opening is spherical because that requires a minimum increase in its surface energy. To determine the size, we calculate the net vertical force due to the surface tension T when the radius of the drop is R . When this force becomes smaller than the weight of the drop, the drop gets detached from the dropper. **[IIT-JEE-2010]**

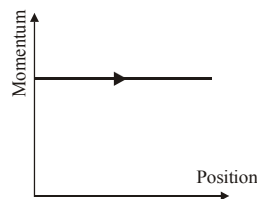
8. If the radius of the opening of the dropper is r , the vertical force due to the surface tension on the drop of radius R (assuming $r \ll R$) is

(A) $2\pi rT$ (B) $2\pi RT$ (C) $\frac{2\pi r^2 T}{R}$ (D) $\frac{2\pi R^2 T}{r}$

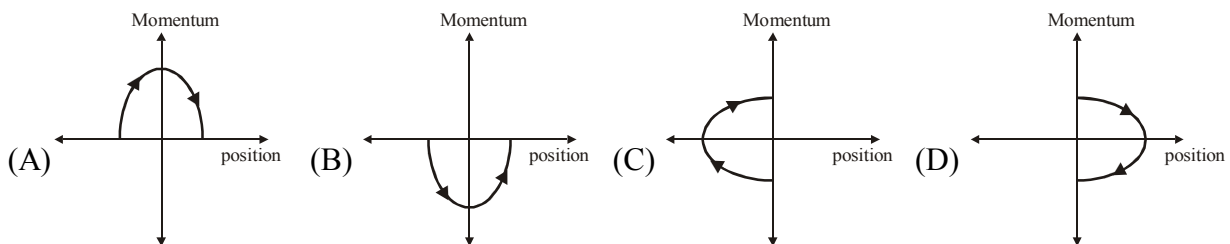
9. If $r = 5 \times 10^{-4} \text{ m}$, $\rho = 10^3 \text{ kg m}^{-3}$, $g = 10 \text{ ms}^{-2}$, $T = 0.11 \text{ Nm}^{-1}$, the radius of the drop when it detaches from the dropper is approximately
 (A) $1.4 \times 10^{-3} \text{ m}$ (B) $3.3 \times 10^{-3} \text{ m}$ (C) $2.0 \times 10^{-3} \text{ m}$ (D) $4.1 \times 10^{-3} \text{ m}$
10. After the drop detaches, its surface energy is
 (A) $1.4 \times 10^{-6} \text{ J}$ (B) $2.7 \times 10^{-6} \text{ J}$ (C) $5.4 \times 10^{-6} \text{ J}$ (D) $8.1 \times 10^{-6} \text{ J}$

Paragraph for Questions Nos. 11 to 13

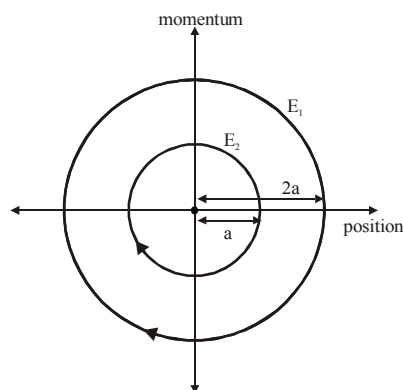
Phase space diagrams are useful tools in analyzing all kinds of dynamical problems. They are especially useful in studying the changes in motion as initial position and momentum are changed. Here we consider some simple dynamical systems in one-dimension. For such systems, phase space is a plane in which position is plotted along horizontal axis and momentum is plotted along vertical axis. The phase space diagram is $x(t)$ vs. $p(t)$ curve in this plane. The arrow on the curve indicates the time flow. For example, the phase space diagram for a particle moving with constant velocity is a straight line as shown in the figure. We use the sign convention in which position or momentum upwards (or to right) is positive and downwards (or to left) is negative. [IIT-JEE 2011]



11. The phase space diagram for a ball thrown vertically up from ground is

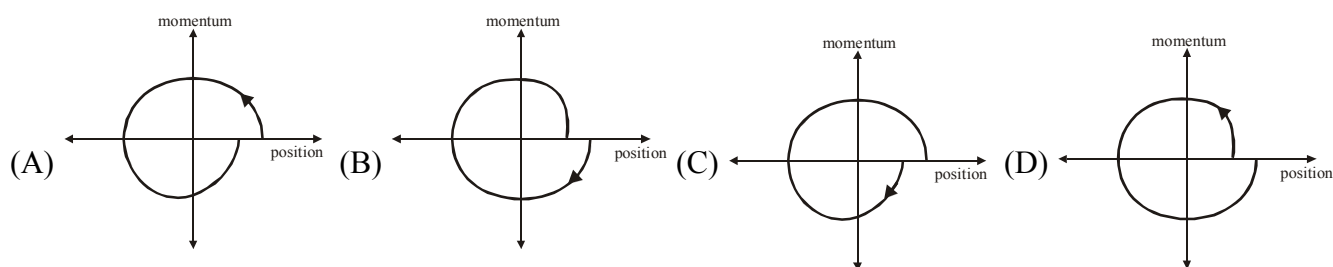
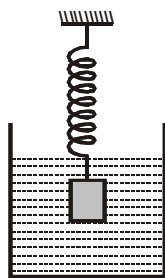


12. The phase space diagram for simple harmonic motion is a circle centered at the origin. In the figure, the two circles represent the same oscillator but for different initial conditions, and E_1 and E_2 are the total mechanical energies respectively. Then

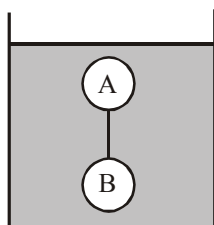


- (A) $E_1 = \sqrt{2}E_2$ (B) $E_1 = 2E_2$ (C) $E_1 = 4E_2$ (D) $E_1 = 16E_2$

13. Consider the spring-mass system, with the mass submerged in water, as shown in the figure. The phase space diagram for one cycle of this system is



14. Two solid spheres A and B of equal volumes but of different densities d_A and d_B are connected by a string. They are fully immersed in a fluid of density d_F . They get arranged into an equilibrium state as shown in the figure with a tension in the string. The arrangement is possible only if [IIT-JEE-2011]



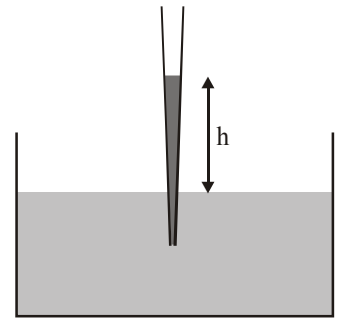
- (A) $d_A < d_F$ (B) $d_B > d_F$ (C) $d_A > d_F$ (D) $d_A + d_B = 2d_F$
15. A thin uniform cylindrical shell, closed at both ends, is partially filled with water. It is floating vertically in water in half-submerged state. If ρ_c is the relative density of the material of the shell with respect to water, then the correct statement is that the shell is [IIT-JEE-2012]
- (A) more than half-filled if ρ_c is less than 0.5
 (B) more than half-filled if ρ_c is more than 1.0
 (C) half-filled if ρ_c is more than 0.5
 (D) less than half-filled if ρ_c is less than 0.5
16. A solid sphere of radius R and density ρ is attached to one end of a mass-less spring of force constant k . The other end of the spring is connected to another solid sphere of radius R and density 3ρ . The complete arrangement is placed in a liquid of density 2ρ and is allowed to reach equilibrium. The correct statement(s) is (are) [IIT-JEE-2013]

- (A) the net elongation of the spring is $\frac{4\pi R^3 \rho g}{3k}$ (B) the net elongation of the spring is $\frac{8\pi R^3 \rho g}{3k}$
 (C) the light sphere is partially submerged. (D) the light sphere is completely submerged.

17. A glass capillary tube is of the shape of truncated cone with an apex angle α so that its two ends have cross sections of different radii. When dipped in water vertically, water rises in it to height h , where the radius of its cross section is b . If the surface tension of water is S , its density is ρ , and its contact angle with glass is θ , the value of h will be (g is the acceleration due to gravity)

[JEE Advanced-2014]

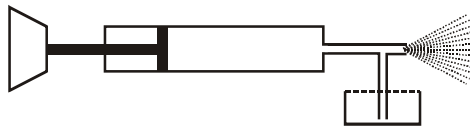
- (A) $\frac{2S}{b\rho g} \cos(\theta - \alpha)$ (B) $\frac{2S}{b\rho g} \cos(\theta + \alpha)$
 (C) $\frac{2S}{b\rho g} \cos(\theta - \alpha/2)$ (D) $\frac{2S}{b\rho g} \cos(\theta + \alpha/2)$



Paragraph for Questions 18 to 20

A spray gun is shown in the figure where a piston pushes air out of a nozzle. A thin tube of uniform cross section is connected to the nozzle. The other end of the tube is in a small liquid container. As the piston pushes air through the nozzle, the liquid from the container rises into the nozzle and is sprayed out. For the spray gun shown, the radii of the piston and the nozzle are 20 mm and 1 mm respectively. The upper end of the container is open to the atmosphere.

[JEE Advanced-2014]



18. If the piston is pushed at a speed of 5 mms^{-1} , the air comes out of the nozzle with a speed of
 (A) 0.1 ms^{-1} (B) 1 ms^{-1} (C) 2 ms^{-1} (D) 8 ms^{-1}
19. If the density of air is ρ_a and that of the liquid ρ_ℓ , then for a given piston speed the rate (volume per unit time) at which the liquid is sprayed will be proportional to

- (A) $\sqrt{\frac{\rho_a}{\rho_\ell}}$ (B) $\sqrt{\rho_a \rho_\ell}$ (C) $\sqrt{\frac{\rho_\ell}{\rho_a}}$ (D) ρ_ℓ

20. A person in a lift is holding a water jar, which has a small hole at the lower end of its side. When the lift is at rest, the water jet coming out of the hole hits the floor of the lift at a distance d of 1.2 m from the person. In the following, state of the lift's motion is given in List I and the distance where the water jet hits the floor of the lift is given in List II. Match the statements from List I with those in List II and select the correct answer using the code given below the lists.

[JEE Advanced-2014]

List - I

- (P) Lift is accelerating vertically up.
 (Q) Lift is accelerating vertically down with an acceleration less than the gravitational acceleration.
 (R) Lift is moving vertically up with constant speed.
 (S) Lift is falling freely.

List-II

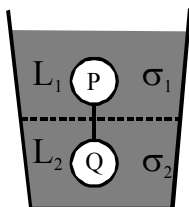
- (1) $d = 1.2 \text{ m}$
 (2) $d > 1.2 \text{ m}$
 (3) $d < 1.2 \text{ m}$
 (4) No water leaks out of the jar

Code :

- (A) P-2, Q-3, R-2, S-4 (B) P-2, Q-3, R-1, S-4
 (C) P-1, Q-1, R-1, S-4 (D) P-2, Q-3, R-1, S-1

21. Two spheres P and Q of equal radii have densities ρ_1 and ρ_2 , respectively. The spheres are connected by a massless string and placed in liquids L_1 and L_2 of densities σ_1 and σ_2 and viscosities η_1 and η_2 , respectively. They float in equilibrium with the sphere P in L_1 and sphere Q in L_2 and the string being taut (see figure). If sphere P alone in L_2 has terminal velocity $\vec{V}_P$ and Q alone in L_1 has terminal velocity $\vec{V}_Q$, then

[JEE Advanced-2015]



- (A) $\frac{|\vec{V}_P|}{|\vec{V}_Q|} = \frac{\eta_1}{\eta_2}$ (B) $\frac{|\vec{V}_P|}{|\vec{V}_Q|} = \frac{\eta_2}{\eta_1}$ (C) $\vec{V}_P \cdot \vec{V}_Q > 0$ (D) $\vec{V}_P \cdot \vec{V}_Q < 0$
22. Consider two solid spheres P and Q each of density 8 gm cm^{-3} and diameters 1 cm and 0.5 cm , respectively. Sphere P is dropped into a liquid of density 0.8 gm cm^{-3} and viscosity $\eta = 3 \text{ poise}$. Sphere Q is dropped into a liquid of density 1.6 gm cm^{-3} and viscosity $\eta = 2 \text{ poise}$. The ratio of the terminal velocities of P and Q is.
- [JEE Advanced-2016]
23. A drop of liquid of radius $R = 10^{-2} \text{ m}$ having surface tension $S = \frac{0.1}{4\pi} \text{ Nm}^{-1}$ divides itself into K identical drops. In this process the total change in the surface energy $\Delta U = 10^{-3} \text{ J}$. If $K = 10^\alpha$ then the value of α is
- [JEE Advanced-2017]
24. A uniform capillary tube of inner radius r is dipped vertically into a beaker filled with water. The water rises to a height h in the capillary tube above the water surface in the beaker. The surface tension of water is σ . The angle of contact between water and the wall of the capillary tube is θ . Ignore the mass of water in the meniscus. Which of the following statements is (are) true?
- [JEE Advanced-2018]
- (A) For a given material of the capillary tube, h decreases with increase in r
 (B) For a given material of the capillary tube, h is independent of σ .
 (C) If this experiment is performed in a lift going up with a constant acceleration, then h decreases.,
 (D) h is proportional to contact angle θ .
25. Consider a thin square plate floating on a viscous liquid in a large tank. The height h of the liquid in the tank is much less than the width of the tank. The floating plate is pulled horizontally with a constant velocity u_0 . Which of the following statements is (are) true ?
- [JEE Advanced-2018]
- (A) The resistive force of liquid on the plate is inversely proportional to h
 (B) The resistive force of liquid on the plate is independent of the area of the plate
 (C) The tangential (shear) stress on the floor of the tank increases with u_0 .
 (D) The tangential (shear) stress on the plate varies linearly with the viscosity η of the liquid.

ANSWER KEY

EXERCISE (S-1)

1. Ans. 45° , $9600\sqrt{2}$ Pa 2. Ans. $\frac{L}{2}\rho(g+a)$ 3. Ans. $P_0 - \rho a \ell$ 4. Ans. 37.5 N
5. Ans. 8 6. Ans. 2 7. Ans. $H = \frac{L^2\omega^2}{2g}$ 8. Ans. $\frac{5d}{4}$
9. Ans. (a) 5, (b) $\frac{2}{3}$ 10. Ans. $f = \frac{1}{2\pi}\sqrt{\frac{3g}{2R}}$ 11. Ans. 45° 12. Ans. nH
13. Ans. 4 14. Ans. 2.52 cm 15. Ans. $h = \frac{mg + 4sa}{\rho_w a^2 g}$ 16. Ans. 20 N
17. Ans. 4.9 litre/min 18. Ans. $6.43 \times 10^{-4} \text{ m}^3/\text{s}$ 19. Ans. 5 cm
20. Ans. 11 m/s 21. Ans. $\frac{h_2 + h_1}{2}$ 22. Ans. 108 23. Ans. 20.4 m
24. Ans. 5 s 25. Ans. 5 26. Ans. $\frac{dQ}{dt} \propto r^5$

EXERCISE (S-2)

1. Ans. 2.79 gm/cc 2. Ans. $\omega = \sqrt{\frac{3g}{2L}\left(\frac{d_2 - d_1}{d_1}\right)}$ 3. Ans. 1.9
4. Ans. $R\sqrt{\frac{3}{2}}$ 5. Ans. 2 6. Ans. $h = \frac{2m}{\pi}$, $H = \frac{11}{32\pi} \text{ m}$
7. Ans. 5 8. Ans. $x = 2 \text{ m}$ 9. Ans. $h_2 = 3 h_1$
10. Ans. $+29625 \text{ J/m}^3$, -30000 J/m^3 11. Ans. 4 12. Ans. 8
13. Ans. (a) $\left[\frac{dgRH^2}{3}\right]$, (b) $\frac{dg\pi R^2 H}{6}$ 14. Ans. $\left(\frac{4g}{H}\right)x$ 15. Ans. 0.1 cm
16. Ans. 3 : 9 : 5 17. Ans. $\sqrt{\frac{18g}{19a}}$ 18. Ans. 4.5m 19. Ans. $\left(\frac{R}{24d}\right)^{\frac{1}{3}}$
20. Ans. 8 21. Ans. $\frac{\lambda a}{2y} g$ 22. Ans. 2
23. Ans. $v = \sqrt{2g\left[h_1 + h_2\left(\frac{\rho_2}{\rho_1}\right)\right]}$ 24. Ans. 5
25. Ans. (a) 10 m/s, (b) 2.5 m/s, (c) 4.688 m, (d) $3.14 \times 10^{-3} \text{ m}^3/\text{s}$ 26. Ans. 6
27. Ans. 4

EXERCISE (O-1)

- | | | | | | |
|-------------------------------------|---|----------------|----------------|--------------|--------------|
| 1. Ans. (C) | 2. Ans. (C) | 3. Ans. (C) | 4. Ans. (C) | 5. Ans. (A) | 6. Ans. (D) |
| 7. Ans. (C) | 8. Ans. (C) | 9. Ans. (D) | 10. Ans. (D) | 11. Ans. (A) | 12. Ans. (B) |
| 13. Ans. (B) | 14. Ans. (D) | 15. Ans. (D) | 16. Ans. (A) | 17. Ans. (B) | 18. Ans. (D) |
| 19. Ans. (A) | 20. Ans. (B) | 21. Ans. (B) | 22. Ans. (A) | 23. Ans. (B) | 24. Ans. (A) |
| 25. Ans. (C) | 26. Ans. (B) | 27. Ans. (D) | 28. Ans. (C) | 29. Ans. (C) | 30. Ans. (C) |
| 31. Ans. (C) | 32. Ans. (A) | 33. Ans. (D) | 34. Ans. (B) | 35. Ans. (B) | 36. Ans. (C) |
| 37. Ans. (A) | 38. Ans. (C) | 39. Ans. (D) | 40. Ans. (B) | 41. Ans. (B) | 42. Ans. (B) |
| 43. Ans. (A) | 44. Ans. (B) | 45. Ans. (D) | 46. Ans. (A) | 47. Ans. (A) | 48. Ans. (C) |
| 49. Ans. (A,D) | 50. Ans. (B, C) | 51. Ans. (A,C) | 52. Ans. (A,C) | | |
| 53. Ans. (A)-R; (B)-Q; (C)-Q; (D)-Q | 54. Ans. (A)-P,T; (B)-P,S; (C)-Q,T; (D)-R | | | | |

EXERCISE (O-2)

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|-------------------------------------|---|--------------|--------------|--------------|--------------------|
| 1. Ans. (D) | 2. Ans. (D) | 3. Ans. (A) | 4. Ans. (B) | 5. Ans. (A) | 6. Ans. (A) |
| 7. Ans. (C) | 8. Ans. (A) | 9. Ans. (B) | 10. Ans. (B) | 11. Ans. (B) | 12. Ans. (D) |
| 13. Ans. (D) | 14. Ans. (D) | 15. Ans. (C) | 16. Ans. (D) | 17. Ans. (D) | 18. Ans. (A) |
| 19. Ans. (D) | 20. Ans. (D) | 21. Ans. (B) | 22. Ans. (D) | 23. Ans. (D) | 24. Ans. (D) |
| 25. Ans. (D) | 26. Ans. (C) | 27. Ans. (D) | 28. Ans. (C) | 29. Ans. (C) | 30. Ans. (B, C, D) |
| 31. Ans. (C) | 32. Ans. (B) | 33. Ans. (A) | 34. Ans. (A) | 35. Ans. (C) | 36. Ans. (C) |
| 37. Ans. (A)-Q, (B)-R, (C)-P, (D)-S | 38. Ans. (A)-P,T; (B)-Q,R,S; (C)-Q,R,S,T; (D)-P | | | | |

EXERCISE (JM)

- | | | | | | |
|--------------|--------------|-------------|--------------|--------------|--------------|
| 1. Ans. (1) | 2. Ans. (4) | 3. Ans. (3) | 4. Ans. (3) | 5. Ans. (2) | 6. Ans. (2) |
| 7. Ans. (4) | 8. Ans. (2) | 9. Ans. (2) | 10. Ans. (1) | 11. Ans. (3) | 12. Ans. (4) |
| 13. Ans. (3) | 14. Ans. (1) | | | | |

EXERCISE (JA)

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|------------------|------------------|-----------------|-----------------|--------------|----------------|
| 1. Ans. (A) | 2. Ans. (B) | 3. Ans. (D) | 4. Ans. (B) | 5. Ans. (B) | 6. Ans. 6 |
| 7. Ans. 6 | 8. Ans. (C) | 9. Ans. (A) | 10. Ans. (B) | 11. Ans. (D) | 12. Ans. (C) |
| 13. Ans. (B) | 14. Ans. (A,B,D) | 15. Ans. (A) | 16. Ans. (A, D) | 17. Ans. (D) | 18. Ans. (C) |
| 19. Ans. (A) | 20. Ans. (C) | 21. Ans. (A, D) | 22. Ans. 3 | 23. Ans. 6 | 24. Ans. (A,C) |
| 25. Ans. (A,C,D) | | | | | |